

Home Production in the City ^{*}

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Abstract

By 2050, two thirds of the world’s population is predicted to live in cities, predominantly mega-cities. Agglomeration drives up productivity, but the commuting costs may act as a brake by incentivizing specialization within the household, and thus potentially decreasing female labor force participation. Using origin-destination travel data from the mega city of São Paulo in Brazil, I document that labor force participation declines sharply with distance from the city center, a spatial gradient largely driven by married women. When women work, they not only face lower wages, but also commute in slower modes of transport, relying more often on public transport and less on driving than men. To quantify the implications of these patterns for the economy, I model the trade-off between benefits of agglomeration and the cost of commuting using a quantitative spatial framework in which couples and singles decide where to live and whether and where to work. My model reveals that if women faced the same commuting costs as men, labor force participation would increase by 10.3 percentage points—an effect larger than equalizing labor market returns by gender—with especially strong impacts for married women on the urban periphery. Investing in transportation infrastructure that makes commuting equal by gender could have large aggregate and welfare implications.

Keywords: Commuting, Gender, Labor Force Participation

JEL Classification: J16, J22, R13, R23, R541, O18

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1 Introduction

By 2050, over two-thirds of the world’s population is expected to live in cities (United Nations, 2018). The concentration of people and firms in urban areas has long been recognized as a major source of productivity gains, as agglomeration economies raise output and wages through dense labor markets and externalities (Rosenthal and Strange, 2004; Glaeser and Gottlieb, 2009; Combes et al., 2012; Baum-Snow et al., 2024). Yet urban density also increases congestion and commuting costs, which may limit the extent to which individuals benefit from these productivity advantages. If commuting costs incentivize specialization within the household, density may impair female employment, lowering aggregate labor force participation. This, in turn, can have broader implications for urban labor markets and aggregate welfare. In this paper, I study whether commuting affects within-city gender gaps in labor supply and its aggregate and distributional consequences.

There are two main forces through which commuting may act as a barrier to female employment. The first is by affecting the effective return to market work: when commuting costs are high, the net benefit of working outside the home falls, making home production relatively more attractive.¹ The second arises from differences in commuting costs: women often face higher mobility constraints, such as limited access to safe or fast transport. If it is more costly for women than for men to move around the city, home production is even more attractive to them than market work.

I study this in the context of the mega city of São Paulo, Latin America’s largest city. It generates roughly one-tenth of Brazil’s GDP but also has among the longest average commuting times by public transport in the world (see Appendix Figure A1).^{2,3} By 2010, it was home to over 11 million inhabitants, many of whom are out of the labor market. The city offers an example of the coexistence of high productivity and large spatial inequalities in access to jobs. Moreover, the city’s Origin-Destination Surveys (*Pesquisa Origem e Destino*) are ideal for understanding intra-household behavior, as they

¹Throughout the paper, I refer to home production as the alternative to market work. I do not consider home production in the intensive margin; rather, only in the extensive margin, such as in Hsieh et al. (2019) and Caliendo et al. (2019).

²The municipality of São Paulo accounted for 10.3% of Brazil’s GDP in 2019. Source: <https://agenciadenoticias.ibge.gov.br/en/agencia-news/2184-news-agency/news/32610-sao-paulo-accounted-for-10.3-of-brazilian-gdp-in-2019>.

³According to the Moovit Public Transport Index (2022), São Paulo ranks 10th in the world, with an average travel time by public transport of 62 minutes. Source: <https://moovitapp.com/insights/en/Moovit-Insights-Public-Transit-Index-cities>.

provide detailed individual, household, and trip information for over 125,000 people, 43,000 households, and 250,000 trips.

Using these surveys, I document several stylized facts on labor supply and commuting by gender and marital status. Labor supply declines sharply with distance from the city center, as average commuting times increase. This pattern likely reflects sorting by income, as individuals outside the labor market tend to live in cheaper neighborhoods, with lower access. Controlling for individual characteristics such as wealth quintile and education largely eliminates this spatial gradient for singles and married men.⁴ However, it remains pronounced for married women, consistent with household specialization.

I then examine differences in commuting outcomes among those who work. Using reported travel times, distances and mode of transportation from the survey, I find that, controlling for individual characteristics, there are no statistically significant differences in the commuting time reported by single and married men and married women, despite married women covering significantly shorter distances in their commutes. The pattern is largely explained by transport mode: women, both single and married, are more reliant on public transport, which is slower and less flexible. This suggests that married women work closer to home not necessarily because they value proximity more, but because traveling the same distance is effectively more costly for them. Single women, for whom time constraints are less binding, are willing to commute farther and longer, despite having similar car usage rates.

Labor market returns also differ systematically by gender and marital status. Even after controlling for occupation, sector, and education, large wage gaps remain between men and women. I interpret these residual wage wedges as arising from discrimination in the labor market, which further lowers the effective return to market work for women, particularly among those facing higher commuting costs.⁵

What are the costs to the economy resulting from these gendered commuting frictions? Understanding their aggregate costs requires a framework in which men and women trade off the benefits of agglomeration with the costs of commuting, allowing for endogenous labor supply. It should also incorporate how firms and individuals com-

⁴Throughout the text, I use “married” and “in a couple” interchangeably to refer to heterosexual couples living together, whether formally married or not. All other working age individuals are considered single. See Appendix Section A.2 for more details.

⁵These differences could reflect variations in productivity or discrimination, but it is not possible to distinguish between the two with the data.

pete for floor space in the city, and how prices, both wages and rents, respond to these demands.

To formalize these mechanisms, I develop a quantitative spatial model in which men and women, single or in couples, inhabit a city and choose where to live and whether and where to work. They consider the wages, commuting costs, housing prices, and amenities associated with their location and labor supply choices. The model allows for endogenous labor force participation and within-couple specialization, meaning that one partner may optimally withdraw from the labor market when commuting costs are high. In this sense, commuting costs act as a force for specialization: because couples take into account both partners' commuting costs, it can be less costly for the household if one member stays at home while the other works.

I structurally estimate the model to quantify how gender differences in commuting costs and labor market discrimination shape labor supply and spatial sorting within the city. The model does not take a stand on why women are less likely to use private modes of transport, and it considers the mode of transport choice as exogenous. Under this assumption, it considers type specific travel time matrices, that consider the frequency at which these individuals use each mode as I observe in the data.⁶

With the travel time matrices at hand, I estimate gravity equations for couples and singles to recover some of the models parameters. Importantly, the gravity equation for couples can be estimated by considering the commuting time for husband and wife simultaneously, under the assumption that commuting costs enter household utility in a multiplicative way. This functional form grants tractability to estimate the model and fits the patterns present in the data very well. The results of the gravity equation indicate that married men are less sensitive to commuting time than all other types, with smaller semi elasticities of commuting flows with respect to time in absolute value.

The estimation proceeds by recovering location-workplace choice probabilities from the data using observed residence and workplace locations by gender and marital status. In the model, home production is treated as another workplace. I then invert the model and recover the fundamental parameters, such as productivity and amenities. Workplace amenities and home production amenities in the model can be interpreted as gender norms, which predict why some choices may be chosen more despite not seeming economically attractive.

⁶Tsivanidis (2022) and Pietrabissa (2023) compute the travel time matrices for high-skilled and low-skilled using the same approach.

I conduct two accounting counterfactual exercises to assess how gender differences affect the city. In the first, I remove labor market discrimination by equalizing men's and women's wages at each location. The increased return to market work induces an aggregate increase in labor supply of 9 p.p., from a baseline in which 23.2% of the city's population are out of the labor market. The effects are mostly driven by married women moving from home production to market work. In contrast, married men increase their rates of home production due to specialization forces within couples.

In the second counterfactual, I equalize commuting conditions between men and women, allowing them to face the same travel times and commuting sensitivities. This would decrease home production rates in the city by 10.3 percentage points, a slightly larger increase relative to equalizing wages. Unlike the previous counterfactual, removing commuting differences is relatively more effective at decreasing home production in the periphery, with large impacts on the spatial gradient of home production in the city.

While both counterfactuals raise aggregate output through higher labor supply, the increase is larger when commuting differences are removed. Reducing spatial frictions allows individuals to access the most productive areas of the city, amplifying the productivity gains from greater participation. By contrast, welfare gains are more muted when wages are equalized; higher nominal incomes boost housing demand, especially in central locations, driving up floor space prices and offsetting part of the real income gains from eliminating discrimination. Equalizing commuting costs, in turn, delivers much larger welfare improvements by directly reducing travel disutility and flattening the housing price gradient, enabling households to reach good jobs without paying for central area's housing prices.

Importantly, equalizing commuting conditions is not necessarily equivalent to improving transport infrastructure. Expanding public transport networks or reducing congestion may shorten average travel times, but such improvements do not necessarily close gender gaps in commuting if women's commutes remain slower than men's. Policies that improve infrastructure without addressing these gender-specific barriers may therefore have limited effects on labor market participation and aggregate output.

One important caveat is that the counterfactual results are likely an upper bound on the actual effects. The exercise assumes that all individuals respond identically to changes in commuting and wages, which is unlikely to hold in reality. Nonetheless, the direction of the bias is the same in both counterfactuals, so the comparison between the

two remains informative.

Finally, in light of the two counterfactual results, while the two counterfactuals are simply accounting the impacts of gender differences in the economy, reducing differences in commuting between men and women seems like a more tangible outcome than fully eliminating discrimination in the labor market. Yet, in the model, both policies yield comparable gains in participation and aggregate output. This suggests that targeted interventions that reduce gendered commuting frictions can be a powerful tool to promote growth in cities.

Related literature A large literature documents gender differences in both the willingness to commute and realized commuting patterns. Women tend to accept shorter commutes and value commute reductions more than men, trading off wages for proximity (White, 1986; Le Barbanchon et al., 2021; Petrongolo and Ronchi, 2020; Borghorst et al., 2024; Bütikofer et al., 2025).⁷ Much of this literature treats commuting time and distance as interchangeable measures of commuting costs. Using data that jointly report commuting time, distance, and main mode of transport, I show that this assumption is misleading.⁸

A second strand of work emphasizes the effects of barriers to women’s mobility, often emphasizing safety and monetary costs. I show that differences in access to faster modes of transport can impact outcomes of women. Prior studies highlight the role of monetary costs (Dasgupta and Datta, 2023; Chen et al., 2024) and safety concerns (Borker et al., 2021; Garlick et al., 2025) as key obstacles. These constraints shape women’s educational outcomes (Cheema et al., 2019; Borker et al., 2021; Fiala et al., 2022; Alba-Vivar, 2024), employers hiring behavior (Buchmann et al., 2024), and women’s job-switching decisions (Sharma, 2023). In Mexico City and São Paulo, women exhibit a positive willingness to pay for women-only subway cars, consistent with safety risks raising commuting costs (Aguilar et al., 2021; Kondylis et al., 2025), and Christensen and Osman (2023) find that discounted Uber rides in Cairo increase female mobility, largely through safety improvements.⁹

⁷Oreffice and Sansone (2023) show that gender commuting differences are present mostly in same-sex couples, suggesting that gender-conforming norms play an important role.

⁸Le Barbanchon et al. (2021) show that the gender gap in reservation wages and maximum acceptable commute varies by geography and suggest that car dependence in some locations may be a key explanation.

⁹For a thorough review of the literature on barriers to female autonomy and labor market participation, see Heath et al. (2024) and Jayachandran (2020).

A third and closely related literature examines how commuting frictions shape women’s labor market outcomes. Much of this work studies cross-city differences in female labor force participation in the US (Black et al., 2014; Moreno-Maldonado, 2022; Farré et al., 2023). Evidence from China, a rapidly urbanizing economy, also shows that rising travel times explain a large share of the recent decline in women’s labor force participation (Liu et al., 2025). This paper is closer to research that explores this relationship within cities (Kawabata and Abe, 2018; Liu and Su, 2022; Velásquez, 2022; Ranosova, 2025).¹⁰ The contribution is to embed these mechanisms in a general equilibrium framework with endogenous location and labor supply decisions. Through counterfactual analysis, I study how removing gender differences in commuting affects choices between home production and market work. In practice, this could occur through improved transportation infrastructure, as shown in studies finding positive effects of new transport links on women’s labor supply (Seki and Yamada, 2020; Velásquez, 2022; Caldwell and Danieli, 2024).^{11,12} An important caveat, however, is that infrastructure improvements alone do not guarantee gender-neutral commuting conditions, as emphasized in the counterfactual exercises in this paper.

This paper builds on the classic theories of time allocation and household specialization developed by Becker (1965) and Mincer (1962), in which the returns to market work depend on wages and participation costs. High commuting costs lower the effective return to market work, reinforcing specialization: the partner with higher potential wages commutes to work while the other stays home to produce without incurring those costs. Carta and De Philippis (2018) and Ranosova (2025) also link commuting to household specialization. My paper builds on and extends these papers by considering a general equilibrium setting, capturing how changes in labor supply feed back into agglomeration and local prices.

A large literature quantifies the aggregate output losses from gender inequality. Declining barriers in occupational choice and human-capital accumulation account for a substantial share of U.S. growth (Hsieh et al., 2019), and related work finds sizable productivity gains from narrowing gender gaps in participation and entrepreneurship (Cav-

¹⁰Joshi (2024) shows that commuting affects women’s choice between agricultural and non-agricultural jobs.

¹¹Infrastructure effects are not always positive for women; Bütikofer et al. (2024) show that gaining access to a larger labor market increased the gender wage gap.

¹²Relatedly, other spatial policies that increase female labor supply include housing (Franklin, 2020), childcare (Hojman and Boo, 2022), and working from home (Ho et al., 2024; Jalota and Ho, 2024).

alcanti and Tavares, 2016; Cuberes and Teignier, 2016). Ngai and Petrongolo (2017) show that the marketization of home production increases aggregate output by reallocating women’s time toward market activities, while Bell et al. (2019) document large innovation losses from unequal access to opportunity. My paper shows how gendered spatial frictions may also impact aggregate outputs.

My paper also relates to the literature on quantitative spatial models, which provide structural frameworks to study how spatial frictions shape the organization of economic activity across space. Most models are designed to explain the spatial distribution of *paid* economic activity, where firms locate and where workers supply market labor, while abstracting from the geography of *unpaid* economic activity, such as home production. Canonical contributions formalize how productivity, amenities, and commuting costs shape the equilibrium allocation of employment and income across space (Allen and Arkolakis, 2014; Ahlfeldt et al., 2015; Monte et al., 2018), and subsequent work has systematized these insights into general frameworks for quantifying urban structure and welfare (Redding and Rossi-Hansberg, 2017; Redding, 2023, 2024). I extend the general framework by introducing home production as an alternative to market work, endogenizing labor supply decisions, in a similar fashion to Hsieh et al. (2019) and Caliendo et al. (2019).

Yet, to understand the spatial distribution of home production and non-market work, it is necessary to model decisions made within the household. In my model, couples choose location and labor supply choices simultaneously. A growing set of papers embeds couples into quantitative spatial frameworks to study joint residential and workplace decisions within cities (Velásquez, 2022; Ranosova, 2025), showing that commuting costs and workplace heterogeneity generate meaningful gender and welfare implications.¹³ Relative to these papers, I allow for endogenous labor supply choices for all individuals, so that commuting frictions influence whether and where individuals work. This extension links the city’s spatial structure directly to the extensive margin of labor supply and to the distribution of home production across neighborhoods. Furthermore, I consider that commuting times of men and women may differ due to access to faster modes of transport, considering another way in which commuting can be gendered.

The role of household structure in location choices has received increasing attention in the quantitative spatial literature. Across cities, Fan and Zou (2021) integrate local

¹³Moreno-Maldonado (2022) develops a quantitative framework in which couples decide whether to reside in small or large cities.

labor and marriage markets to show how the joint search for jobs and partners shapes the spatial distribution of skills, while [Moreno-Maldonado \(2022\)](#) considers how city size interacts with female labor supply choices. Within cities, [Moreno-Maldonado and Santamaria \(2022\)](#) considers how delayed fertility influenced the urban revival in US cities, and [Ahlfeldt et al. \(2025\)](#) study how family type and age impact residential sorting within the city of Copenhagen.

The remainder of the paper is structured as follows. Section 2 describes the data and context of the mega-city of São Paulo. Section 3 presents the theoretical framework, building on the patterns present in the stylized facts. Section 4 describes the estimation of the model. Section 5 presents the counterfactual results, where I quantify the aggregate costs of gendered commuting. Section 6 concludes and proposes the next steps for this research agenda.

2 Data and Context

In this section, I first describe the data sources used for the empirical evidence and for the structural estimation. I follow by describing the context of the city. Finally, I show empirical support for the forces in the model by documenting facts about employment and labor supply by gender and marital status in the city.

2.1 Data

In my empirical and quantitative analysis, I combine data from different sources including household surveys, administrative records, and the national census. I next describe the two main data sources and briefly describe where I obtain additional variables.

Household Travel Surveys The main data source are the household travel surveys for the São Paulo Metropolitan Region for the years 2007, and 2017. Each cross section contains a random sample of households, representative at a small spatial level. The surveys contain information at the household level, describing the household composition, and individual characteristics of all the members, such as education level and activity status. The surveys report the coordinates of residence, workplace, and educational locations, when existing. It also indicates the type of school, sector of work, and occupation.

The survey provides detailed information on the daily trips of 126,046 individuals, living in 43,419 households. They report all the trips taken by each household member in the previous day, which is necessarily a weekday. For each trip, the data reports the

mode of transport, motive of the trip at origin and destination, reported travel time, with time of departure and time of arrival. Therefore, for each member of the household, I know their activities and measures of mobility.

Travel time matrix To obtain the travel times by mode of transport for each I create a travel time matrix by different modes of transport using a realistic routing algorithm (r5r) for the year 2007 (Pereira et al., 2021). Data from Open Street Maps provide the surface roads’ speed and direction. These are used by the algorithm to calculate a distribution of travel time by car and walking between any two points in the city. For travel times by public transit, in addition to road surface data, the algorithm requires General Transit Feed Specifications (GTFS) datasets, which provide the schedule, routes, and stops of the public transportation network of the city. This way, it is possible to compute the travel time by public transport between locations. The earliest available GTFS dataset is from 2012, and I create a counterfactual 2007 version by removing transport links that did not exist then. Appendix Section A.4 validates these matrices against reported trip durations from the Origin–Destination survey and shows that the resulting bias is symmetric across gender–marital-status groups.

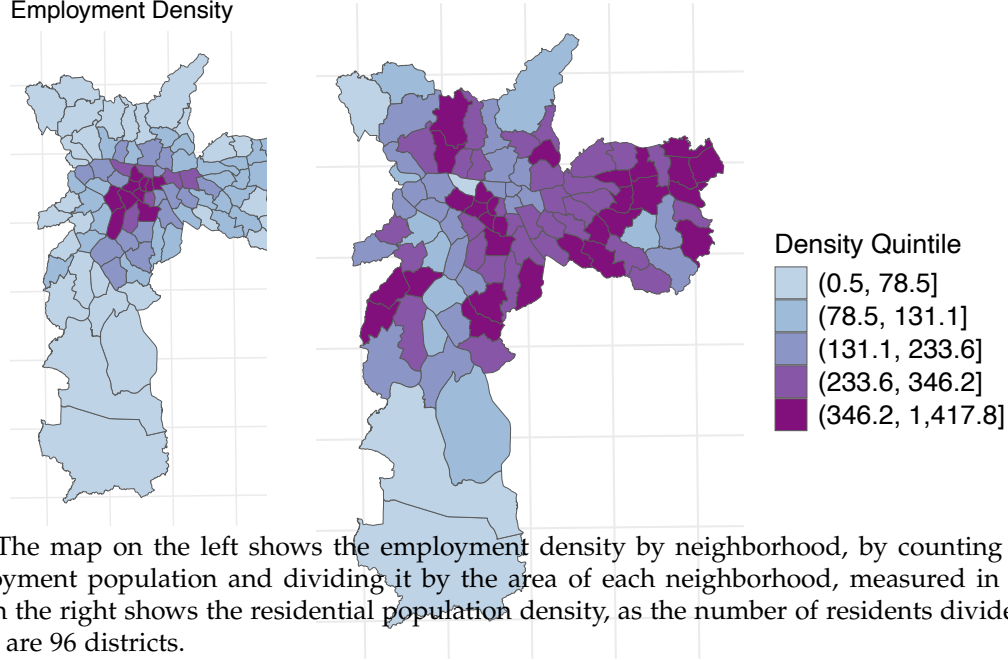
Additional data I use the National Budget Survey (*Pesquisa do Orçamento Familiar*) to compute the housing expenditure shares, and the Property Tax Registry (ITBI) to obtain floor space prices for both residential and commercial floor space. See Appendix Section A for additional details on the datasets.

2.2 Context

The city of São Paulo is the largest city in Latin America in terms of population, with more than 11 million inhabitants, and serves as Brazil’s primary economic center. The municipality alone accounts for roughly one tenth of the country’s GDP and concentrates a large share of formal employment, offering a diverse set of job opportunities. Despite this economic importance, it is a city characterized by severe spatial frictions, resulting from a rapid and largely unplanned urbanization process. The average reported commuting time in 2007 was close to 54 minutes, and commutes that rely on public transportation as the main mode are substantially longer on average. Many regions of the city remain underserved in terms of infrastructure and services, both privately and publicly provided, which limits accessibility to jobs.

Figure 1 maps the employment and population density quintiles, using the same

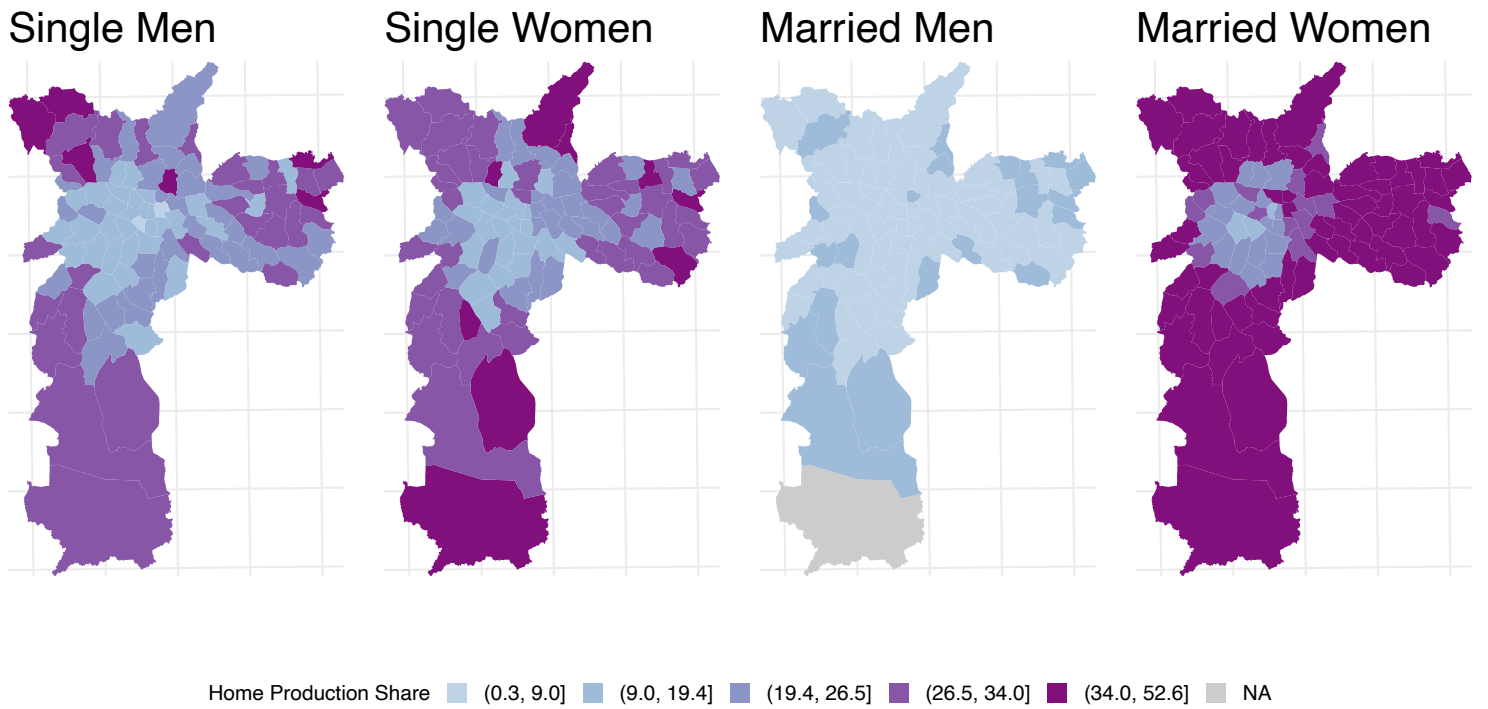
Figure 1: Employment and Population Density in São Paulo



scale. They show that while just a few central neighborhoods concentrate most employment in the city, the same does not occur for residential choices. Population is much more spread out across different areas of the city, with many peripheral neighborhoods exhibiting large concentration of residents.¹⁴ This is consistent with other cities in the world, in which most employment and jobs agglomerate in just a few locations. The center business district is chosen as the Se station, a historical landmark in the city center, and its location is highlighted in Appendix Figure A2.

Furthermore, a large share of the working age population of the city is out of the labour market and gender gaps in participation are pronounced. Appendix Table A1 shows an almost 20 percentage point difference between men and women who report working. This gap is almost entirely explained by differences in women reporting being house wives. The other activities, not working, retired and student, report close to no gender differences. Throughout the paper, I pool together those who report not working with house wives into the home production category. Throughout the remainder of the paper, the sample does not include retirees or students, keeping only those that are working or in home production.

¹⁴The extreme south and north of the city have presence of environmentally protected areas, and for that reason have much lower population density.



Notes: The figure maps the share of home production of working-age individuals engaged in home production across neighborhoods of São Paulo, separately for married women, married men, single women, and single men. Darker shares indicate higher shares of home production. Source: Origin-destination survey, 2007.

2.3 Stylized Facts

To motivate the quantitative analysis, this subsection documents a set of stylized facts that describe the spatial organization of market and non-market activity and the gendered transport frictions within the city of São Paulo. Together, these patterns highlight the patterns that the model seeks to explain.

Figure 2 maps the share of working-age individuals not engaged in market work across neighborhoods, by gender and marital status. The prevalence of home production varies significantly across space.¹⁵ However, this distribution is mediated by gender and marital status. Three main patterns emerge. First, the spatial distribution of home production for singles, men or women, is relatively similar. Second, among couples, we see a clear pattern of specialization, with married men rarely doing home production, and their wives exhibiting home production shares of over 34% across most of the city. Finally, for all groups, labor supply is markedly higher in the city center, where job opportunities are and transport costs are lower.

The spatial patterns in Figure 2 are partly driven by observables: income and social inequality shape residential sorting in the city (Harari, 2024), and the rent gradient pushes

¹⁵In the city there are 96 neighborhoods or districts, which are used for most of the empirical analysis. In the model estimation, I use a more aggregated spatial unit which divides the city into 23 zones.

households with lower resources toward peripheral neighborhoods. To characterize how strongly the location–home–production relationship varies across groups, and to assess how much of it survives standard controls, I regress the probability of being in home production on distance to the CBD for singles and couples, with and without individual and household controls. The exercise is descriptive: I report the conditional association between location and labor force participation, not the causal effect of moving an individual farther from the center.

Column (1) of Table 1 shows that a 10km increase in distance from the center is associated with a 5 percentage point higher probability of being out of the labor market for single men, and an 8 percentage point higher probability for single women. Column (2) shows that the gradient is considerably smaller for married men relative to all other individuals, while married women face a seven percentage point increase, similar to single women. In the distance to CBD distribution, 10 km is approximately the difference between the 25th and 75th percentiles.

Columns (3) and (4) of Table 1 report the same gradients controlling for observable characteristics, such as age, household composition, education level and a measure of wealth quintiles.¹⁶ Adding controls eliminates the gradient for single men and halves it for single women, with the remaining coefficient only marginally significant. For couples the picture is different: the differential gradient on married women is essentially unchanged in size and significance, and the gradient for married men flips sign. The latter pattern is consistent with within-couple specialization—husbands are more likely to work *where* their wives are in home production, and vice versa—and it persists for the wealthiest households and for couples without children.

Distance to the CBD is a coarse summary of accessibility: a 10 km radius captures very different effective commutes depending on where in the city one lives. The mechanism that matters for the analysis in the rest of the paper is the actual cost of traveling from a residence to the bulk of the city’s employment, which depends on travel time rather than straight-line distance. Appendix Table A4 therefore replicates the same regressions replacing distance to the CBD with the log of the average commute time observed at the residence district. The qualitative pattern is unchanged: the unconditional gradient is positive for all groups and steeper for women, controls eliminate it for single

¹⁶I compute the wealth quintiles based on reported assets in the household, not on earnings. This approach has been used in the DHS surveys, and applied to other survey and census datasets (Smits and Steendijk, 2015; Zipfel et al., 2022; Bandiera et al., 2022).

Table 1: Probability of Home Production by Distance to the CBD

	P(Home Production)			
	Singles (1)	Couples (2)	Singles (3)	Couples (4)
Distance to CBD (10km)	0.05*** (0.01)	0.02*** (0.007)	0.01 (0.01)	-0.01* (0.007)
Distance to CBD (10km) $\times$ Female	0.03** (0.02)	0.05*** (0.01)	0.03* (0.02)	0.05*** (0.01)
Observations	16,229	16,541	16,229	16,541
Individual-level controls			✓	✓
Household-level controls			✓	✓
Education FEs			✓	✓
Wealth quintile FEs			✓	✓

Notes: Robust standard errors in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: SP origin destination survey for 2007. Sample includes adults aged 18 to 55 living in the city of São Paulo who are working or in home production. Controls include age, age squared, household size, number of 0-3 year-olds, 4-10 year olds and number of 75+ in the household in the household. Fixed effects for education level and for wealth quintile based on household assets are included.

men and roughly halve it for single women, and the differential for married women remains positive, large, and statistically significant. The relationship between residential accessibility and home production therefore does not hinge on the choice to measure accessibility through distance to the CBD; it survives when accessibility is measured in the units—travel time—that map directly into the rest of the analysis. These findings are not causal but others have found causal effects of increased commuting times on female labor supply (Farré et al., 2023; Liu et al., 2025).

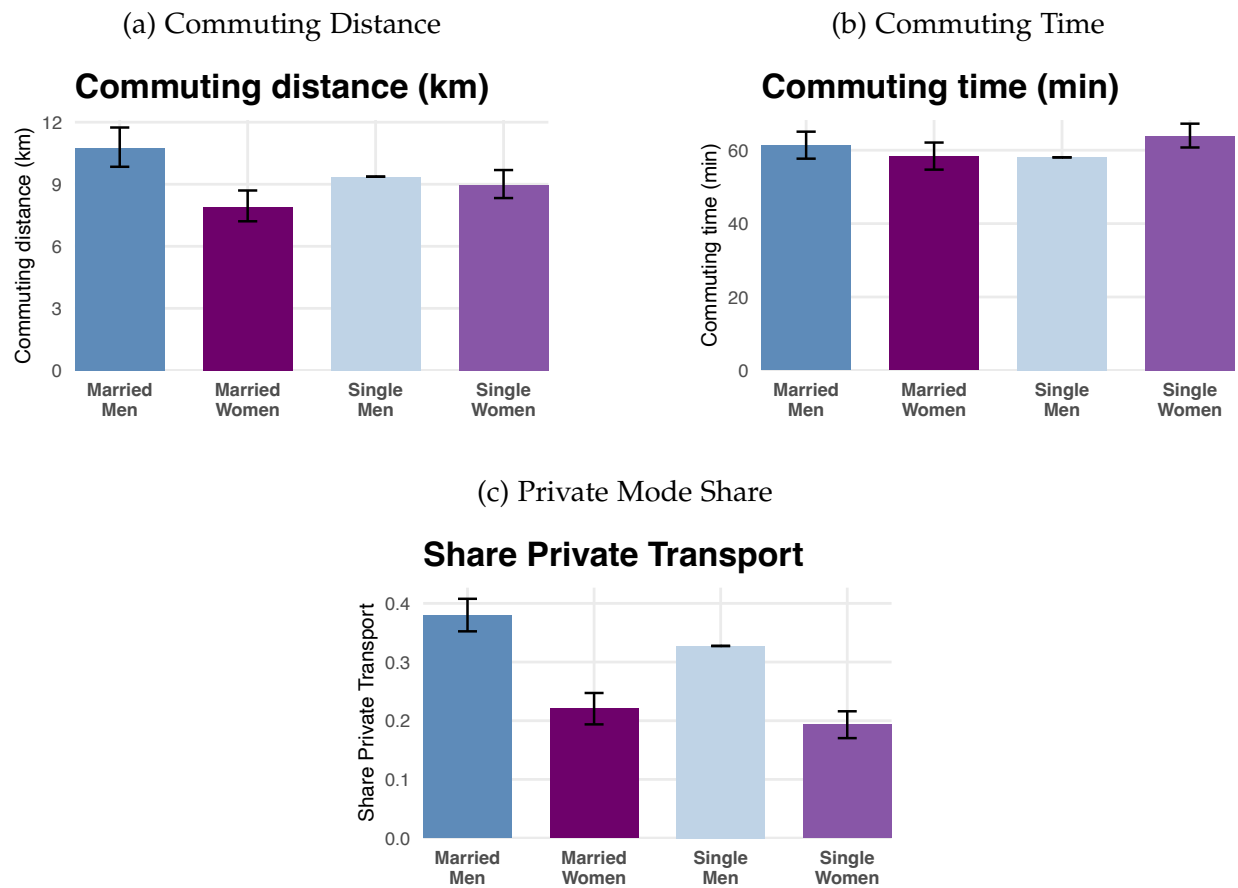
The evidence above points to clear spatial and gender differences in who participates in the labor market. I next look at whether there are differences in outcomes when individuals participate in the labor market. First, I look at how commuting behavior varies across groups, and I run regressions of commuting outcomes on gender and marital status indicators, controlling for individual characteristics such as age, education, household composition, and neighborhood fixed effects. The results of the regression are in Appendix Table A5 and the outcomes of interest are commuting distance, time and whether the main mode of transport is private. Using the effect sizes from the regressions, I calculate the adjusted means relative to the baseline average of the baseline group (single men).

Figure 3a shows the differences in adjusted means, and shows that while married women work 7.9 km away from home on average, married men work 10.8 km away from home, 39% more distant. In contrast, the average commuting distance of single men and women are statistically equal, with 9.4 km and 9 km respectively. These stark differences in commuting distance are not reflected in commuting times. Figure 3b shows that commuting times are long for all groups, with a one-way commute lasting close to 60 minutes. Only single women present an average time statistically different from single men, with commutes lasting 64 minutes. One potential reason why women are commuting the same or longer than men, despite working relatively closer to home, is about *how* they commute.

Figure 3c plots the share who commute to work using a private mode of transport. Women are much less likely to use these modes than men, regardless of marital status. Across many cities of the world using public transit is slower than commuting by car. The share of the area where travel time favors public transportation over car use is estimated to be less than 1% in São Paulo (Liao et al., 2020). Access to a private mode of transport has a clear impact on how fast individuals can go around the city and, consequently, the distances that they are able to cover. Both married and single women are much less likely to use private transport, but single women still travel similar distances to single men by accepting longer commuting times, unlike married women. This pattern alone, however, does not allow me to distinguish between two possible mechanisms: whether men travel farther because they drive and can access more distant jobs, or whether women travel shorter distances because they rely on slower transport modes and therefore face tighter spatial constraints on where they can work. In the model, the commuting cost depends on time between locations, but also how sensitive individuals are to increases in commuting time. This allows for married and single women to face similar travel time between locations, but have different willingness to commute longer.

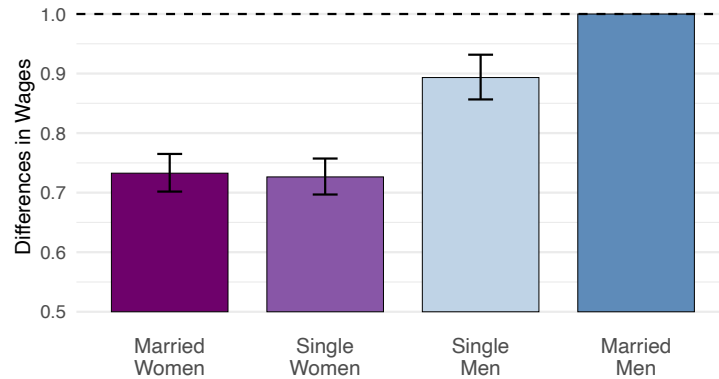
Finally, Figure 4 document differences in labor market returns across groups. This is the first channel traditionally emphasized in the literature to explain within-household specialization (Becker, 1965; Mincer, 1962). The higher-earning spouse focused on market work, and the other would allocate more time to home production. After controlling for hours of work, sector, occupation, and education level, married men still earn more than observationally equivalent married women, single women, and single men. While the labor market returns for women are very similar, single men earn less than married

Figure 3: Differences in Commuting Outcomes by Gender and Marital Status



Notes: The figure plots the predicted commuting outcomes (distance, time, and private mode share) for each demographic group, calculated as the baseline mean for single men plus the estimated regression effects for each group. The adjusted means control for age, education, wealth, household composition, and residence location bin fixed effects. The confidence intervals are also derived from the regression estimates (see Appendix Table A5).

Figure 4: Different Returns in the Labor Market by Gender and Marital Status



Notes: The figure shows earnings differences across demographic groups, estimated from a log earnings regression controlling for age, education, sector, occupation, job type (between regular and flexible), and hours worked on the previous day. The dashed line indicates the reference group (married men) and error bars show 95% confidence intervals. See Appendix Table A6.

men, but still significantly more than women.¹⁷

While these gaps could also reflect differences in other unobservable characteristics, such as talent, they could also occur due to discriminatory behavior by firms. It is not possible to disentangle these factors, and I choose to interpret them as discrimination, as in [Hsieh et al. \(2019\)](#), as there is no reason to believe that women are systematically less talented than men. The presence of a marital premium for men is consistent with specialization allowing them to devote more time and effort to market work, therefore increasing their market returns.

3 A Quantitative Spatial Model with Home Production

I develop a quantitative urban model which accounts for the spatial sorting of families in space and for their labor supply choices. In this model, men and women inhabit the city, and they may live in households as single or as couples.¹⁸ The city is formed by different neighborhoods that will vary in terms of their productivity in the production of a tradable good, and in terms of amenities. Amenities make neighborhoods more or less attractive to live in. Neighborhoods can also vary in terms of their home-production pull, which allows the benefit of staying out of the labor market to vary across space. This would capture for instance different gender norms around the city, or access to

¹⁷This is consistent with evidence in the US showing that there still exist a marriage premium for men, there is no longer a marriage penalty for women [Juhn and McCue \(2017\)](#)

¹⁸The model only considers different-sex couples, and there is no endogenous formation of couples.

amenities that are complementary to home production. Importantly, individuals face different travel times across neighborhoods, which in the model shows up as different travel time matrices, constructed taking the transportation modes as given.

3.1 Model Setup

I consider a city with a discrete set of locations $\Theta = \{1, \dots, N\}$, which in my data are the finite set of neighborhoods. There are $N + 1$ workplaces $\Theta_E = \{0, 1, \dots, N\}$, where workplace indexed by zero indicates home production. The city is populated by men and women indexed by $g \in \{m, f\}$, respectively. Households are indexed by $o = \{s, c\}$ after singles and couples. Therefore, there are four types of individuals, and three types of households. All households choose residence and workplace(s) simultaneously. For couples, there is a trilateral choice for the residence indexed by n , the husband's workplace indexed by i , and the wife's workplace indexed by j . Single households choose residence and one workplace location as in [Ahlfeldt et al. \(2015\)](#). Individuals og may differ in terms of their commuting cost semi-elasticity, their travel time matrix, and the returns they obtain from market work.

I consider a closed city model, with a fixed number of individuals in each group L^{og} . Locations differ in terms of productivity in the good sector A_n , and amenities B_{nij}^{og} . Firms use labor and floor space to produce a market good m_{nij} , which is the numeraire in the economy with $P = 1$. Individuals out of the labor market are equally productive in producing home goods, which are perfectly substitutable with market goods, and are consumed within the household. They also receive a transfer from the government (e.g. unemployment insurance or cash transfers). All markets are competitive. For conciseness, the model is derived only for couples here, see Appendix Section B for the extended version adding singles.

3.2 Preferences

Consider a household c with residence in location n , with the husband's workplace in location i and the wife's workplace in location j . Let $\phi_{ij} \in \{0, 1, 2\}$ indicate how many household members are in home production. The utility of household ω depends on consumption of goods c_{nij} and the consumption of residential floor space h_{nij} , location amenities B_{nij}^c , and commuting costs from residence n to workplaces i and j , d_{ni}^{cm} and

d_{nj}^{cf} .¹⁹ Preferences are Cobb-Douglas and couples spend a share $1 - \beta^c$ of their income on floor space:

$$U_{nij}^c(\omega) = \frac{B_{nij}^c z_{nij}^c(\omega)}{d_{ni}^{cm} d_{nj}^{cf}} \left(\frac{c_{nij}}{\beta^c} \right)^{\beta^c} \left(\frac{h_{nij}}{1 - \beta^c} \right)^{1 - \beta^c}, \quad \beta^c < 1, \quad (1)$$

where the iceberg commuting cost is defined as $d_{ns}^{cg} = e^{\kappa^{cg} t_{ns}^{cg}} \in [1, \infty)$. Travel time between residence n and workplace s is denoted by t_{ns}^{cg} . This formulation implies commuting costs that rise exponentially with travel time. The parameter κ^{cg} controls the rate at which commuting costs increase, and is allowed to vary across individuals, capturing heterogeneous sensitivity to travel time.

Households draw a residence-workplace specific idiosyncratic shock $z_{nij}^c(\omega)$, which captures unobserved determinants of the joint choice of residence n , the husband's workplace i , and the wife's workplace j . The home sector is modeled as an additional potential workplace location, representing home production.

Individuals face different returns in the labor market, earning different wages when working for firms, and zero earnings otherwise ($w_0^{og} = 0$). The household maximizes utility subject to a budget constraint:

$$m_{nij} + h_{nij} q_{Rn} \leq (1 - \tau)(w_i^{cm} + w_j^{cf}) + \phi_{ij} T \quad (2)$$

where the price of residential floor space is given by q_{Rn} and the price of the market good is $P = 1$. Individuals that work for firms are taxed by the government with a tax rate τ to fund an exogenous transfer T to individuals out of the labor market. This transfer allows households to pay for market goods and housing, even if they don't have wage income. When in home production, individuals produce a quantity φ of home produced goods. Therefore, the total quantity of goods consumed is given by the sum of goods produced by the household and the goods purchased in the market: $c_{nij} = m_{nij} + \varphi \mathbb{1}(j = 0) + \varphi \mathbb{1}(i = 0)$. Perfect substitutability between home and market

¹⁹The location amenity for couples is composed of a residential term, a workplace term for each household member, and a home-production pull for each residence location. The full expression is $B_{nij}^c = B_{Rn}^c B_{Fi}^{cm} B_{Fj}^{cf} B_{n0}^{cm} B_{n0}^{cf}$.

goods (m_{nij}) implies that the shadow price of home goods equals that of market goods.

$$m_{nij} + h_{nij}q_{Rn} \leq (1 - \tau)(w_i^{cm} + w_j^{cf}) + \phi_{ij}T \quad (3)$$

$$c_{nij} + h_{nij}q_{Rn} \leq (1 - \tau)(w_i^{cm} + w_j^{cf}) + \varphi \mathbb{1}(j = 0) + \varphi \mathbb{1}(i = 0) + \phi_{ij}T \quad (4)$$

Implicitly, home produced goods increase the effective budget constraint by allowing the household to increase expenditure by how much they save on goods when producing them at home. The optimization of utility leads to the optimal demands of the goods and housing, conditional on the residential location and the location of workplaces:

$$c_{nij} = \beta^c((1 - \tau)(w_i^{cm} + w_j^{cf}) + \phi_{ij}(\varphi + T)) \quad (5)$$

$$h_{nij} = \frac{1 - \beta^c}{q_{Rn}}((1 - \tau)(w_i^{cm} + w_j^{cf}) + \phi_{ij}(\varphi + T)) \quad (6)$$

Let the household total income be defined as $HI_{ij}^c = (1 - \tau)(w_i^{cm} + w_j^{cf}) + \phi_{ij}(\varphi + T)$. Substituting the demands on the utility function, I obtain the indirect utility function of couples:

$$V_{nij}^c(\omega) = \frac{B_{nij}^c HI_{ij}^c z_{nij}^c(\omega)}{d_{ni}^{cm} d_{nj}^{cf} q_{Rn}^{1 - \beta^c}} \quad (7)$$

which depends on the household total income, the price of floor space, location amenities, labor market participation amenities, commuting costs of household members, and the household idiosyncratic shock. The trade-off between doing working or not is captured in the indirect utility by the different returns and amenities couples receive for each choice. Market work increases household income, but it incurs commuting disutility, unlike the home sector, which has a fixed return and no commuting costs ($d_{n0}^{cg} = 1$). An important feature that allows for specialization in couples, with one person in market work with the other in home sector, is the multiplicative form of the commuting costs of husband and wife. In practice, it augments the agglomeration force for working couples, and it make specialization attractive in the peripheries of cities, where commutes are longer. Finally, I assume that for each couple household, the idiosyncratic

component of utility z_{nij}^c is drawn from an independent Fréchet distribution:

$$F(z) = e^{-z^{-\epsilon^c}}, \quad \epsilon^c > 1, \quad (8)$$

This implies that the household's location and labor supply decisions are made jointly and simultaneously, and are governed by the same elasticity. It seems natural that individuals choose where to live and whether to work or not simultaneously, and that these decisions are not independent.²⁰

3.3 Residence and Workplaces Decision

Couples choose the residence and workplaces to maximize their utility. Since z_{nij}^c are Fréchet-distributed, the share of households in each residence and workplace(s) presents a closed form solution:

$$\pi_{nij}^c = \frac{\left(B_{nij}^c H I_{ij}^c\right)^{\epsilon^c} \left(d_{ni}^{cm} d_{nj}^{cf} q_{Rn}^{1-\beta^c}\right)^{-\epsilon^c}}{\sum_{r=1}^N \sum_{s=0}^N \sum_{p=0}^N \left(B_{rsp}^c H I_{sp}^c\right)^{\epsilon^c} \left(d_{rs}^{cm} d_{rp}^{cf} q_r^{1-\beta^c}\right)^{-\epsilon^c}} \equiv \frac{L_{nij}^c}{L^c} \quad (9)$$

where L_{nij}^c is the measure of couples c living in n and with workplaces i for the husband and j . Therefore, couples sort across residence and workplaces depending characteristics of these location, and their own idiosyncratic preferences.²¹ Summing across all possible workplace pairs, I obtain the probability that a couple may choose to live in location n :

$$\pi_{Rn}^c = \frac{\sum_{s=0}^N \sum_{p=0}^N \left(B_{nsp}^c H I_{sp}^c\right)^{\epsilon^c} \left(d_{ns}^{cm} d_{np}^{cf} q_{Rn}^{1-\beta^c}\right)^{-\epsilon^c}}{\sum_{r=1}^N \sum_{s=0}^N \sum_{p=0}^N \left(B_{rsp}^c H I_{sp}^c\right)^{\epsilon^c} \left(d_{rs}^{cm} d_{rp}^{cf} q_r^{1-\beta^c}\right)^{-\epsilon^c}} \equiv \frac{L_{Rn}^c}{L^c} \quad (10)$$

If I sum across all the possible residence locations, I find the probability that a couple chooses the pair of workplaces ij :

$$\pi_{Fij}^c = \frac{\sum_{r=1}^N \left(B_{rij}^c H I_{ij}^c\right)^{\epsilon^c} \left(d_{ri}^{cm} d_{rj}^{cf} q_r^{1-\beta^c}\right)^{-\epsilon^c}}{\sum_{r=1}^N \sum_{s=0}^N \sum_{p=0}^N \left(B_{rsp}^c H I_{sp}^c\right)^{\epsilon^c} \left(d_{rs}^{cm} d_{rp}^{cf} q_r^{1-\beta^c}\right)^{-\epsilon^c}} \equiv \frac{L_{Fij}^c}{L^c} \quad (11)$$

²⁰This is equivalent to having a nested Fréchet structure, where the three location choice elasticities are the same.

²¹The focus of the model is on the location decision of couples, and I do not add any other worker heterogeneity to the model. Sorting by skill and/or income are also relevant in cities (Tsivanidis, 2022).

Finally, the probability of choosing workplaces i and j conditional on residence location n is:

$$\pi_{nij|n}^c = \frac{\left(B_{n0}^c B_F^c H I_{ij}^c / d_{ni}^{cm} d_{nj}^{cf} \right)^{\epsilon^c}}{\sum_{s=0}^N \sum_{p=0}^N \left(B_{n0}^c B_F^c H I_{sp}^c / d_{ns}^{cm} d_{np}^{cf} \right)^{\epsilon^c}} \equiv \frac{L_{nij}^c}{L_{Rn}^c} \quad (12)$$

and it depends on workplace characteristics, the commuting costs between residence and workplaces and the participation amenity at the residence location.

3.4 Production

Production of Market Goods. All firms operate in perfectly competitive markets and exhibit constant returns to scale. Output price is normalized to one. Firms combine labor and commercial floor space with a Cobb–Douglas technology:

$$Y_t = A_t L_{Ft}^\alpha H_{Ft}^{1-\alpha}, \quad \alpha \in (0, 1), \quad (13)$$

where A_t denotes total factor productivity and α is the labor share. Let q_{Ft} denote the price of commercial floor space.

Employer Utility Maximization. Following [Becker \(2010\)](#), employers may experience taste-based disutility from hiring certain worker groups cg . Let $\partial^{cg} \geq 0$ denote the utility cost per unit of group cg labor. The employer maximizes utility, rather than profits:

$$\max_{L_{Ft}^{cg}, H_{Ft}} U_t^{owner} = A_t L_{Ft}^\alpha H_{Ft}^{1-\alpha} - q_{Ft} H_{Ft} - \sum_{cg} (w_t^{cg} + \partial^{cg}) L_{Ft}^{cg}, \quad (14)$$

where $L_{Ft} = \sum_{cg} L_{Ft}^{cg}$ denotes total employment in location t . The first-order conditions imply for all employed groups cg :

$$\alpha A_t L_{Ft}^{\alpha-1} H_{Ft}^{1-\alpha} = w_t^{cg} + \partial^{cg}, \quad (1 - \alpha) A_t L_{Ft}^\alpha H_{Ft}^{-\alpha} = q_{Ft}, \quad (15)$$

The effective marginal cost of labor is therefore equivalent to the observed wages plus the discrimination cost for the employer.

$$\tilde{w}_t = w_t^{cg} + \partial^{cg}, \quad (16)$$

Therefore, observed wages incorporate the discrimination wedge

$$w_i^{cg} = \tilde{w}_i - \partial^{cg} = (1 - \gamma^{cg}) \tilde{w}_i, \quad \gamma^{cg} = \frac{\partial^{cg}}{\tilde{w}_i}. \quad (17)$$

From (15), cost minimization implies the usual unit-cost condition:

$$\tilde{w}_i = A_i^{1/\alpha} \alpha (1 - \alpha)^{\frac{1-\alpha}{\alpha}} q_{Fi}^{\frac{\alpha-1}{\alpha}}, \quad (18)$$

and the corresponding demand for commercial floor space:

$$H_{Fi} = \left(\frac{(1 - \alpha) A_i}{q_{Fi}} \right)^{1/\alpha} L_{Fi}. \quad (19)$$

Goods market clearing. All firms produce the tradable numeraire good c , whose price is normalized to one. Residents of the city are workers only; firm owners live elsewhere. Local households therefore finance their consumption of c with labor income. Firms sell part of their output within the city to these residents and export the remainder. Exports adjust so that the revenue accruing to firm owners is exactly sufficient to drive owner utility to zero in equilibrium. When discrimination is present, lower wages reduce participation and local demand, and a larger share of output is exported.

Home Produced Goods Individuals in home production produce goods that are consumed within the household, and they all have the same productivity in producing these goods. For any individual who works in $i = 0$, they produce:

$$\varphi = \begin{cases} \varphi, & \text{if } i = 0 \\ 0, & \text{otherwise} \end{cases} \quad (20)$$

Unlike firms, there is no discrimination in home production.

3.5 Commuter Market Clearing

Commuter market clearing requires that the measure of couples that choose workplace pair ij is equal to sum of the measure of couples that live each residence, weighted

by the probability of choosing ij conditional on living in n :

$$L_{Fij}^c = \sum_{n=1}^N \pi_{nij|n}^c L_{Rn}^c \quad (21)$$

This condition also implies that the expected household income for a couple residing in location n equals the sum of wages in both workplaces, across all workplaces, weighted by the conditional probability of commuting:

$$v_n^c = \sum_{s=0}^N \sum_{p=0}^N \pi_{nsp|n}^c HI_{sp}^c \quad (22)$$

3.6 Floor Space Market Clearing

I assume that the prices of commercial and residential floor space are such that the markets clear and supply meets the demand. Under this assumption, the price of commercial floor space is given by the first order condition of the firm profit maximisation problem:

$$q_{Fi} = \frac{(1 - \alpha)}{\alpha} \frac{\tilde{w}_i L_{Fi}}{H_{Fi}} \quad (23)$$

and the price of residential floor space is given the demands for residential floor space from couples within a location:

$$q_{Ri} = \frac{(1 - \beta^c)}{H_{Rn}} v_n^c L_n^c \quad (24)$$

where the demand for residential floor space in a location depends on the expected income of the households, and the measure of households that choose to reside there.²²

3.7 Government Budget Constraint

The government collects taxes from paid workers to fund cash transfers to individuals out of the paid labor market. Every working individual pays a flat tax rate of their wage to the government, which then distributes it among all the individuals that are out of

²²The residential demand is augmented by the singles' demand for floor space when I consider them in the model. This is one of the few changes of adding them, as they compete with the couples for residential floor space.

the labor market. The government's budget constraint is given by:

$$\tau \sum_{g=f,m} \sum_{i=1}^N w_i^{cg} L_{Fi}^{cg} = (L_{F0}^{cm} + L_{F0}^{cf})T \quad (25)$$

3.8 Welfare

Welfare is defined as the sum over the utilities of households over all residence locations:

$$V^c = \left(\sum_{n=1}^N \sum_{i=0}^N \sum_{j=0}^N \left(B_{nij}^c H I_{ij}^c \right)^{\epsilon^c} \left(d_{ni}^{cm} d_{nj}^{cf} q_{Rn}^{1-\beta^c} \right)^{-\epsilon^c} \right)^{1/\epsilon^c} \quad (26)$$

3.9 General Equilibrium

Given the model parameters $\{\alpha, \beta^c, \epsilon^c, \kappa^{cg}, \gamma^{cg}\}$, city population L^c , government cash transfers T , and location characteristics $\{A_i, B_{n0}^c, B_{Ri}^c, B_{Fi}^{cg}, t_{ni}^{cg}\}$, the general equilibrium is represented by the vector of endogenous objects $\{L_{Ri}^c, L_{Fij}^c, \tilde{w}_i, q_{Fi}, q_{Ri}, \tau\}$ determined by the following equations:

1. Residential population equation (10).
2. Labor supply equation (11).
3. Unit cost condition and utility maximization from firms (18).
4. Floor space demand from firms (23).
5. Floor space demand from residences (24).
6. Government budget constraint (25).

3.10 Mechanisms that Lead to Sorting

In the model, three main mechanisms drive the spatial sorting of individuals out of the labor market and toward the periphery of the city. The first and most direct mechanism is that, all else equal, lower-income households tend to locate in cheaper neighborhoods. The second mechanism arises from the fact that individuals who do not participate in the labor market avoid commuting costs and receive a fixed home-production return. For couples living in peripheral areas, where both partners would otherwise face long commutes, this creates strong incentives for specialization, since the household's utility

depends on the commuting costs of both members. Finally, spatial sorting may also be driven by local amenities: neighborhoods in the suburbs may offer higher residential amenities that attract households even when the returns to market work are lower due to longer commutes.

The mechanisms behind the gendered specialization of married women out of market work, while their husbands commute to jobs elsewhere, stem from two key asymmetries. First, men and women face different potential returns in the labor market. Second, they experience different commuting costs, shaped by both travel times across the city and gender-specific sensitivities to commuting disutility (κ^{cg}).²³

4 Model Quantification

In this section, I describe how I estimate the quantitative spatial model using data from the city of São Paulo. The main data sources for the estimation are described in Section 2 and the model is estimated at the zone level, and there are 23 zones in the city. The first two estimation steps recover important parameters in the model. I use wage and workplace data to calculate the efficiency units of labor for each individual type in the economy. I run gravity equations to recover the commuting costs semi-elasticities and the participation amenities for the model individuals. The remaining parameters I calibrate with other data or I borrow from the literature. With the parameters at hand, I invert the model and back out the economy fundamentals, necessary for the counterfactual exercises.

Discrimination Wedges The discrimination wedges are obtained from a log earnings regression at the individual level, controlling for age, age squared, log hours spent at work the previous day, job type, education level and wealth quintile fixed effects, and sector and occupation fixed effects.^{24,25} The regression results are shown in Appendix Table A6 and in Figure 4.

Gravity Equation To estimate the commuting costs elasticities and the home production pull parameters, I rewrite the location choice probability of couples, and apply logs,

²³Gender differences in preferences for home production could also contribute to the observed specialization patterns.

²⁴I classify job types as regular or flexible based on activity status. Regular job accounts for formal employers and employees, civil servants, and sole practitioners, and flexible jobs include informal employers and employees, working in the family business, and self-employed.

²⁵The survey does not contain information on how long individuals have been working, and with education level fixed effects, estimated tenure is collinear with age.

resulting in the following gravity equation:

$$\log(L_{nij}^c) = \underbrace{\eta_{n0}^{cm}}_{B_{n0}^{cm}} + \underbrace{\eta_{n0}^{cf}}_{B_{n0}^{cf}} + \underbrace{\delta_n}_{B_{Rn}^c q_{Rn}^{bc-1}} + \underbrace{\zeta_{ij}}_{B_{Fi}^{cm} B_{Fj}^{cf} H_{ij}^c} + \nu^{cm} t_{ni}^{cm} + \nu^{cf} t_{nj}^{cf} + u_{nij} \quad (27)$$

where η_{n0}^{cm} and η_{n0}^{cf} are fixed effects for residence interacted with husband doing home production and with wife doing home production, respectively. I include residence location fixed effects (δ_n), which absorb the residence amenities and floor space prices. Destination pair fixed effects (ζ_{ij}) absorb the workplace amenities for husband and wife and the household income.²⁶ Travel time between residence and workplaces are given by t_{ni}^{cm} and t_{nj}^{cf} , for the husband and wife, respectively, and they reflect the differences in commuting times that they experience between places due to the different transport modes they use. In Appendix Section A, I describe how I compute the travel time matrices for each individual, and how I validate it with the average reported travel time between locations, available in the origin destination surveys.

The estimated semi-elasticities of commuting are ν^{cm} and ν^{cf} , and they equal $\nu^{c\mathcal{G}} = \epsilon^c \kappa^{c\mathcal{G}}$. The gravity equation for couples increases the dimensionality and sparsity of the flow matrix, as there are many combination of cells that no couple has chosen. I follow by estimating the commuting elasticities using Poisson Pseudo Maximum Likelihood (PPML), following [Silva and Tenreyro \(2006\)](#). Furthermore, there can be endogeneity of travel time with respect to flows, if policy makers improve infrastructure where the demand is already high for example ([Redding, 2024](#)). In order to address this potential endogeneity, I use distance between the origin and destination as instruments for the travel time using the control-function approach, as suggested by [Wooldridge \(2015\)](#).

The estimated origin-home production fixed effects from the gravity equation correspond to the home production pull parameters in the model, which capture how attractive non-market work is in each residential location. Conditional on workplace characteristics, commuting costs, and residential locations, these parameters are the residual component that rationalizes the observed participation rate of married men and women at each origin. Locations where individuals are more likely to be out of the labor market, even after accounting for these observables, must have a stronger home production pull.

Table 2 shows that the commuting costs for married men are smaller than of married

²⁶The fixed effects also absorb the value of home production ($\varphi + T$), which is the return when in home production.

women, indicating that they face lower commuting costs. The absolute value for the semielasticity of married women is 50% larger than of the married men, but they both face relatively small commuting costs, expected due to the long commutes in São Paulo. In Appendix Section X, I show the results of the gravity equation when I consider travel time by car and distance, for both men and women. Using these two other commonly used measures in the gravity equation, they confirm the finding that married women are slightly more sensitive than married men, although with different magnitudes.

Table 2: Gravity equations of married men and women

D.V.	Total Flow (1)
Commuting Time Husband	-0.024*** (0.007)
Commuting Time Wife	-0.036*** (0.007)
Observations	6,788
Residence FE	✓
Workplace Pair FE	✓
Origin-Home Production Wife FE	✓
Origin-Home Production Husband FE	✓

Notes: The table shows the commuting cost semi-elasticities, estimated following the equation (27) with PPML. The unit of observation is residence-workplace of husband-workplace of wife, and the workplaces include home production with travel time equal to zero. Distance between neighborhoods is used as an instrument to travel time in minutes. Own-flows, i.e. commuting within the same neighborhood, are not included in the estimation.

The results of the gravity equations for singles are shown in 3. The first takeaway is that, among singles, the commuting cost parameters are very similar across genders, and close in magnitude to those estimated for married women. This suggests that gender differences in commuting elasticities only emerge within couples, when households internalize both partners' commutes and begin to specialize.

Model Inversion With the gravity coefficients and the estimated home production amenities, I proceed by inverting the model. I can rewrite equation (21) from the commuting market clearing condition as a function of only known parameters and employ-

Table 3: Gravity equations of single men and single women

D.V.	Total Flow	
	(1)	(2)
Commuting Time	-0.033*** (0.005)	-0.036*** (0.005)
Observations	529	529
Residence FE	✓	✓
Workplace FE	✓	✓
Origin-Home Production FE	✓	✓

Notes: The table reports the commuting cost semi-elasticities for single men (column 1) and single women (column 2), estimated using PPML following the same specification as equation (27) but with only one workplace. The unit of observation is residence-workplace, including home production as a potential destination with travel time equal to zero. Distance between neighborhoods is used as an instrument for travel time in minutes. Own-flows (within-neighborhood commuting) are excluded. The origin-home production fixed effects correspond to the estimated home-production pull parameters, B_{n0}^{sg} .

ment and residence data to back out the value of $\omega^c = (B_{Fi}^{cm} B_{Fj}^{cf} H I_{ij}^c)^{\varepsilon^c}$:

$$L_{Fij}^c = \sum_{n=1}^N \frac{B_{n0}^{cm} B_{n0}^{cf} \omega_{ij}^c / \exp(\nu^{cm} t_{ni}^{cm} + \nu^{cf} t_{nj}^{cf})}{\sum_{s=0}^N \sum_{p=0}^N B_{n0}^{cm} B_{n0}^{cf} \omega_{sp}^c / \exp(\nu^{cm} t_{ns}^{cm} + \nu^{cf} t_{np}^{cf})} L_{Rn}^c \quad (28)$$

which is a simple system of equations exactly identified. With the vector of ω_{ij}^c , I follow the procedure proposed by [Ahlfeldt et al. \(2015\)](#), using an algorithm that exploits the variance of wages in the data to as an additional moment. The algorithm searches for the value of ε^c that minimizes the difference between the variance of log wages in the data and the variance of log ω^c . Since I don't observe the monetary equivalent return of doing home production, I use the minimum wage as the equivalent return. I obtain an estimate of $\varepsilon^c = 6.9$ for couples, and lower values for singles, with $\varepsilon^{sf} = 3.57$ for single women and $\varepsilon^{sm} = 3.43$ for single men.

The Fréchet shape parameter governs the dispersion of idiosyncratic preferences in the location-workplace choice. Because couples jointly choose a residence and two workplaces, their choice set is of size $N \times (N + 1) \times (N + 1)$, whereas singles face $N \times (N + 1)$ alternatives. Although the total number of populated cells does not differ drastically—

since the couples represent roughly half of the population, and single men and women each a quarter—the dimensionality of the couples’ problem implies a much larger potential choice set. Consequently, the model requires a higher Fréchet shape parameter to replicate the observed concentration of commuting flows in a relatively small subset of feasible residence-workplace pair combinations. In other words, couples’ decisions appear more deterministic, consistent with the idea that joint household choices exhibit less idiosyncratic dispersion than individual ones.

Data for the average wages for married men at each location are used to construct the effective marginal cost of a unit of labor $\tilde{w}_i$. For individuals subject to firm discrimination, the effective return is adjusted to $(1 - \exp(\gamma^{cf}))\tilde{w}_i$, with γ^{cf} being the coefficients from regression shown in Appendix Table A6. Labor market returns are normalized by the geometric mean so they are in the same scale. With known ω , the Fréchet shape parameter, and the monetary returns at each location, it is possible to back out the workplace amenities. Using the share of commercial floor space, the price of commercial floor space, and the effective marginal cost of labor $\tilde{w}_i$, I recover productivities from the unit-cost condition in Equation (18).²⁷ The residence location amenities for each group are backed out from the residence location choice probability stated in Equation (10). Using the commuting probabilities and wages at each location, I compute the expected household income. Lastly, floor space in each location is estimated from the market clearing condition. Further details of the estimation procedure are in the Appendix Section X.

Model Fit and Parameters With the fundamentals of the model, I solve for the baseline equilibrium to see how well the model does at recovering the endogenous variables that are observed in the data. In this solver, I target the aggregate shares of home production by gender and marital status, but not the location-level home production shares, which remain an untargeted moment in the model. I list the parameters in Table 4, and most are estimated together with the model inversion. The return in home production is set to be the value of the minimum wage in 2007, but the results are robust to using the lowest average wage at any location in the city instead. I impose that the transfer represents 30% of this value, but this assumption does not impact the counterfactuals, as I am not considering changes in the tax schedule. This assumption implies that the labor income tax rate is close to 7%.

²⁷I set the input share of commercial floor space as 20%, consistent with other papers in the literature (Ahlfeldt et al., 2015; Tsivanidis, 2022).

Table 4: Model parameters, values, and sources

Definition	Parameter (value)	Source
<i>Household preferences</i>		
Housing expenditure share (couples)	$\beta^c = 0.20$	Calibrated, Budget Survey
Housing expenditure share (singles)	$\beta^s = 0.25$	Calibrated, Budget Survey
Fréchet shape, couples	$\epsilon^c = 6.9$	Estimated
Fréchet shape, single women	$\epsilon^{sf} = 3.57$	Estimated
Fréchet shape, single men	$\epsilon^{sm} = 3.43$	Estimated
Return in Home Production	$(\varphi + T) = R\$415$	Calibrated
<i>Commuting and mobility</i>		
Commuting cost semi-elasticity	$\nu^{cm} = -0.024$	Estimated
Commuting cost semi-elasticity	$\nu^{cf} = -0.036$	Estimated
Commuting cost semi-elasticity	$\nu^{sm} = -0.036$	Estimated
Commuting cost semi-elasticity	$\nu^{sf} = -0.033$	Estimated
<i>Firms and technology</i>		
Labor share in production	$\alpha = 0.8$	Ahlfeldt et al. (2015)
Taste-based discrimination wedge	γ^{cg}	Estimated
<i>Government and transfers</i>		
Transfer to non-participants	$T = 0.3(\varphi + T)$	Calibrated
Labor income tax rate	$\tau = 0.07$	Estimated

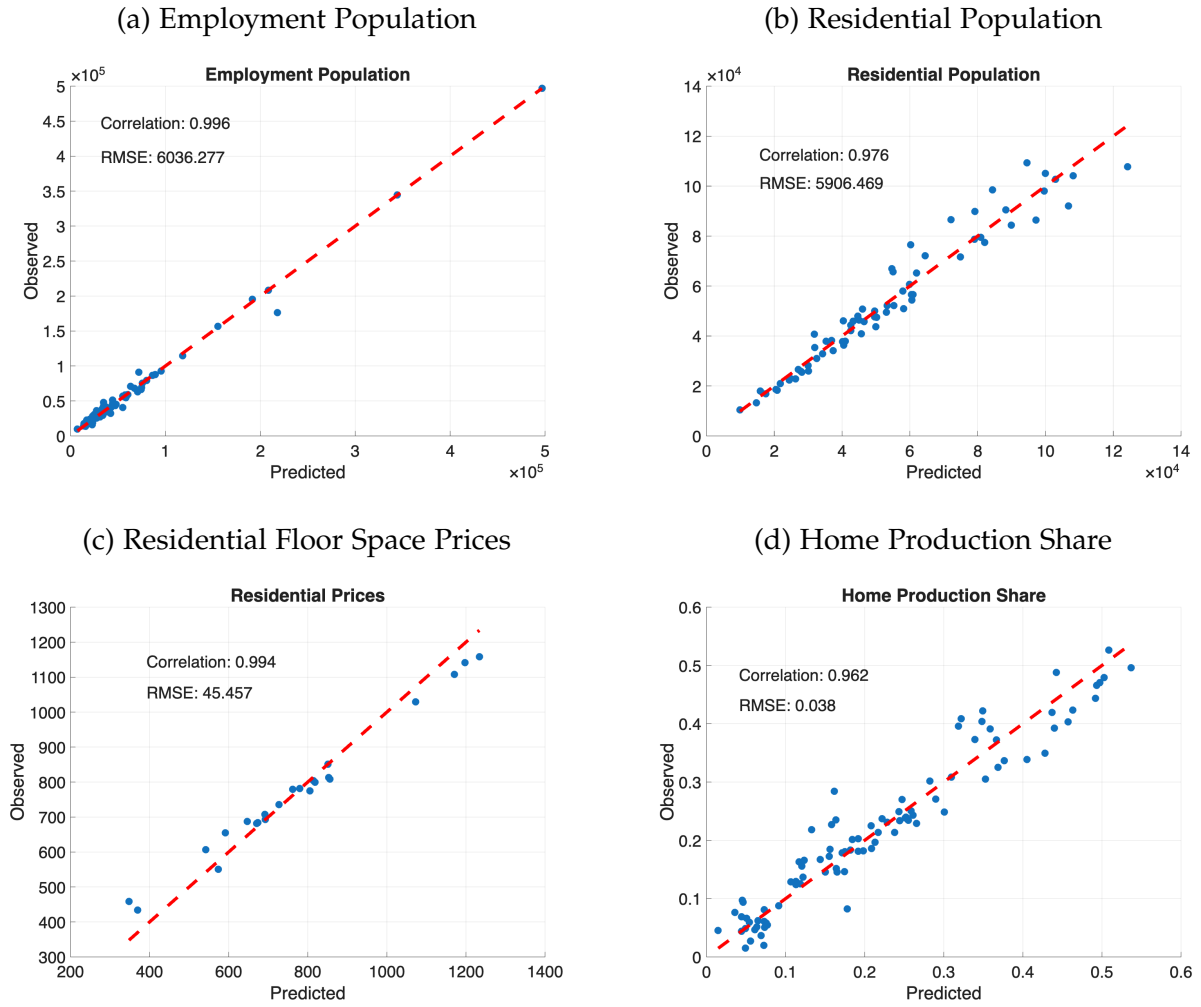
Notes: Estimated parameters are obtained from the gravity equations and model inversion. Calibrated parameters are obtained from data, and/or set exogenously. The remaining parameters are obtained from the literature.

Figure 5 compares the predicted and observed values at the location level for employment population, residential population, the share of households in home production, and residential floor space prices, when I solve for the baseline equilibrium. The model provides a good fit to the observed data across the equilibrium outcomes shown below, and others. The tight clustering of points around the 45-degree line indicates that the estimated fundamentals and parameters reproduce the spatial distribution of paid and unpaid economic activity with accuracy. Even for the share of home production in each location, a moment that is untargeted in the estimation, the model is still able to reproduce quite well the distribution of home production in the city by type. Overall, the correlations between the predictions and the data are very high, over 95% across all variables.

Home production pull A key improvement in the model’s fit comes from explicitly estimating the home production pull parameters, and including it throughout the estimation. This parameter allows the model to capture spatial heterogeneity in labor market participation behavior, which is not explained by commuting costs or other residence specific amenities, which do not interact with home production. Figure 6a shows the relationship between observed and predicted home production shares at each location for married women (CF), when I estimate the model without the home production pull. The model tends to overestimate home production shares in the periphery, and underestimate them in the center of the city. Figure 6b shows that including this parameter in the estimation improves the model’s ability to predict correctly how much home production is done at each location. The correlation increases from 86% to 92%, and the RMSE decreases considerably as well.

This parameter in the data is estimated based on variation in the share of individuals out of the labor market in each location, conditional on commuting times and other residence specific characteristics. Several mechanisms could explain spatial differences in the attractiveness of home production such as childcare accessibility, informal support networks, gender norms, access to other services complementary to home production. Within the model’s framework, the locations with highest home production pulls for women and single men are the most central locations. This is because in these locations, even with great access to good paying jobs and low commuting costs, many people still choose to be out of the labor market. Therefore, the model rationalizes that these are locations that make home production relatively more attractive.

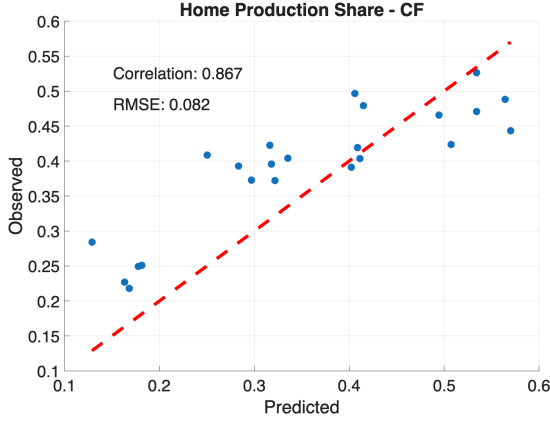
Figure 5: Model Fit



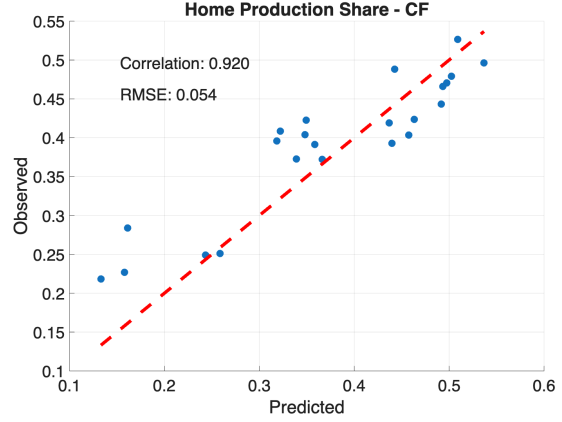
Notes: The figure compares observed and predicted values at the location level for key equilibrium outcomes in the model. Panel (a) plots the employment population, Panel (b) shows the fit for the residential population, Panel (c) displays residential floor space prices and Panel (d) shows the home production share at each location.

Figure 6: Including the Home Production Pull Improves the Model Fit

(a) Share in Home Production without B_{n0}^c



(b) Share in Home Production with B_{n0}^c



Notes: The figure compares observed and predicted values of home production shares of married men under two alternative estimations. Panel (a) shows the relationship between observed and predicted shares when the model is estimated without home production pull. Panel (b) shows the same graph when the model is estimated considering the home production pull.

5 Counterfactual Exercises

I evaluate two counterfactuals using the estimated model, which aim to do an accounting exercise removing the gender differences in the economy. The first removes gender- and marital-status differences in wages, eliminating taste-based discrimination in the labor market and equalizing effective earnings across individuals working in the same location. The second equalizes commuting conditions across groups, removing gender and marital-status differences in both travel times and commuting elasticities. To do so, I set the travel time matrices and commuting cost semi-elasticities equal to those of a married men for all other individuals. In both exercises, I keep the recovered fundamentals fixed and resolve the spatial equilibrium, allowing wages, labor supply, commuting flows, and floor-space prices to adjust endogenously, while holding housing supply constant in each location. This keeps home production amenities fixed, also keeping any potential gender norms fixed.

The model predicts substantial general-equilibrium adjustments under both scenarios, though through different mechanisms. As shown in Figure 7, aggregate output, labor supply, and welfare all increase, but the magnitude and composition of these changes differ across counterfactuals.

When I remove wage differences, firms no longer experience a disutility from hiring

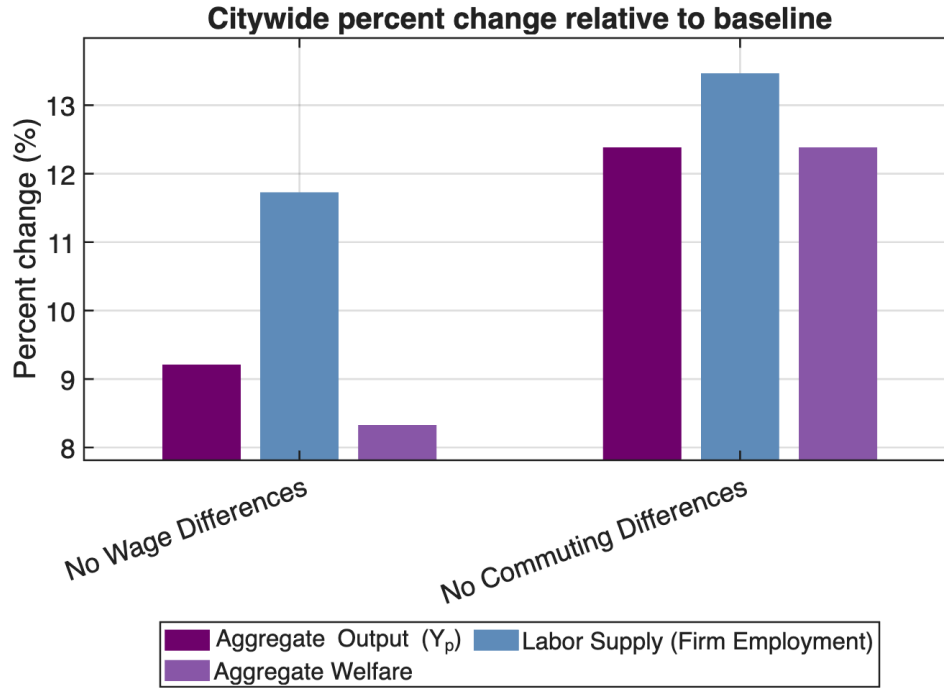
women and single men, and the discrimination wedges that previously lowered their wages disappear. Higher wages raise participation, but commuting frictions remain unchanged. Production exhibits diminishing returns to labor and commercial floor space is fixed, therefore an increase in employment translates into a smaller proportional rise in total output. Workers remain constrained in their spatial choices because of commuting frictions.

When I equalize commuting conditions, aggregate output rises by over 12% and moves more closely with employment. Equalizing travel times and commuting elasticities allows individuals to access more productive central jobs, generating both higher participation and a reallocation of labor toward high-productivity workplaces. Even though nominal wages do not change directly, productivity per worker rises because the spatial organization of employment becomes more efficient. As a result, the city achieves larger aggregate output gains than under the wage-equalization counterfactual, with improvements driven by larger labor supply increases, particularly in very productive locations.

Welfare gains differ sharply across the two counterfactuals. When wage differences are removed, higher nominal incomes increase demand for centrally located housing, putting upward pressure on housing prices and partially offsetting real income gains. In contrast, when commuting conditions are equalized, individuals—especially women in couples—can access better jobs at lower travel costs without large changes in nominal wages or housing demand. Because this counterfactual acts directly on commuting disutility rather than through the income channel, reduces inequality in accessibility without generating large rent-capitalization effects. As a result, the city achieves broader welfare improvements.

Removing wage and commuting differences also reshapes household labor supply patterns. In the baseline, the model reproduces the observed specialization of married couples, with men more likely to work in the market and women more likely to engage in home production, especially in peripheral neighborhoods. Figure 8 shows that under the *No Wage Differences* scenario, home production falls sharply for married women. The rise in their effective market returns weakens the incentive for within-household specialization, as the opportunity cost of staying home increases. Married men's home production rises slightly, reflecting re-optimization within couples once both partners' relative wages are equalized. The effects on labor supply for single women are larger

Figure 7: Counterfactual Changes in Aggregate Outcomes



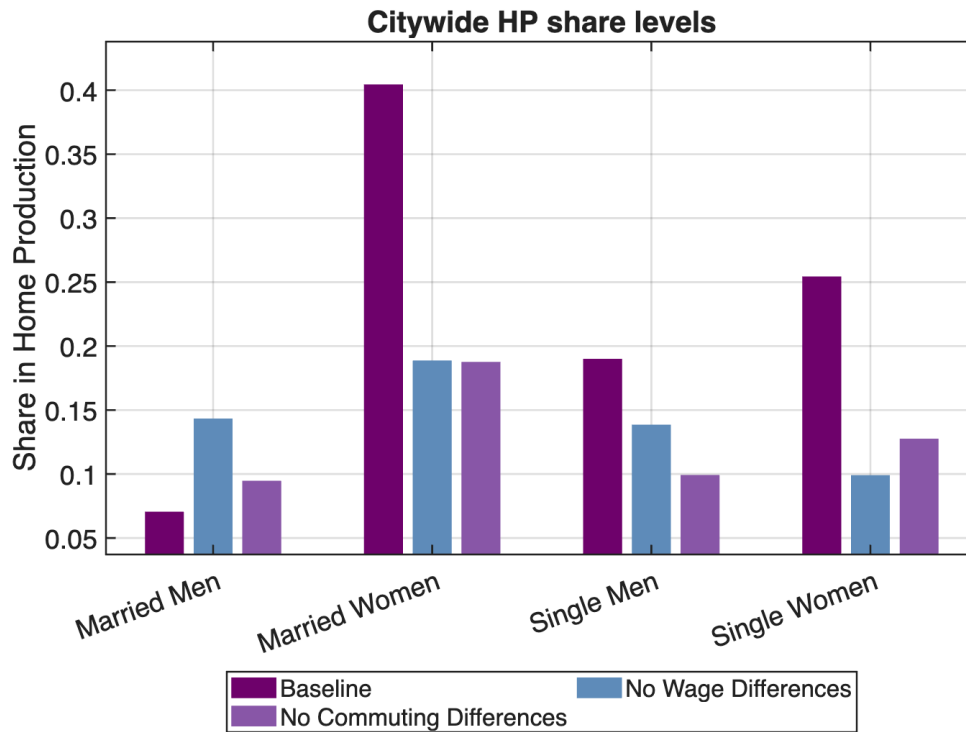
Notes: The bars show the percent change in aggregate output, labor supply, and welfare relative to the baseline level. The first counterfactual equalizes wages between all demographic groups, and the second equalizes commuting costs between all demographic groups.

than for single men because they suffered more discrimination in the baseline.

Under the *No Commuting Differences* scenario, the decline in married women's home production is of a similar magnitude to that observed under wage equalization, but it is accompanied by a smaller increase in married men's home production (Figure 8). This pattern arises because, although commuting costs fall for both partners, the returns to market work remain higher for men, limiting the change in their participation decisions. The reduction in commuting frictions nonetheless lowers within-couple specialization overall, as both partners face a more balanced trade-off between engaging in home production and participating in the labor market. For singles, we observe that reducing commuting frictions has a larger effect for single men than eliminating wage wedges, while for single women the effects are stronger under wage equalization due to the role of discrimination (Figure 8). This pattern suggests that single women's participation is more affected by differential returns in the labor market than single men's, although both groups are substantially constrained by commuting costs.

Both counterfactuals also change the spatial gradient of home production across the

Figure 8: Counterfactual Home Production Shares by Gender and Marital Status

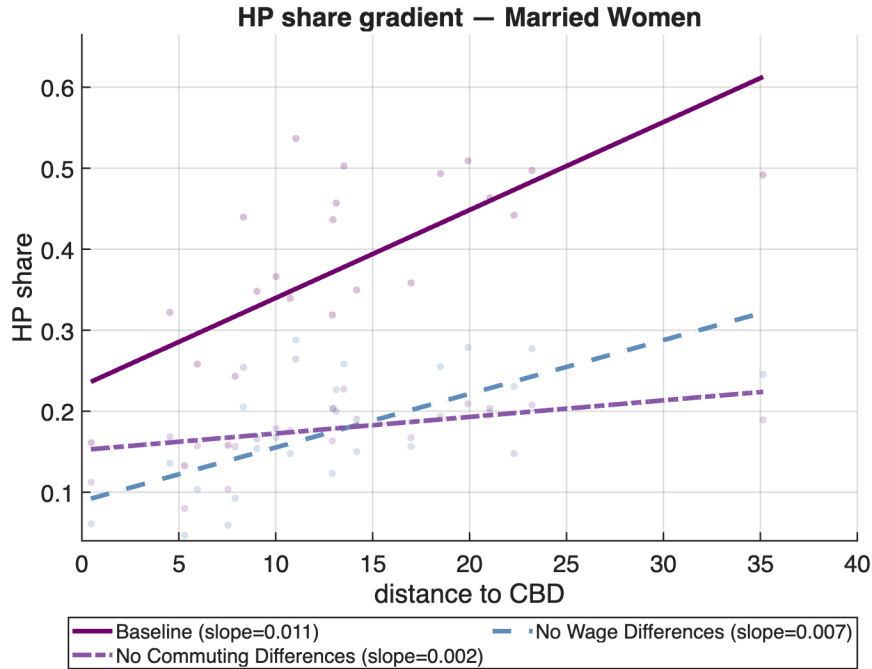


Notes: The figure shows the rates of home production by gender and marital status in baseline and counterfactual scenarios. The dark purple bars show the baseline rates, the blue bars show the counterfactual rates under wage equalization, and the light purple bars show the rates under equal commuting conditions.

city. Figure 9 plots the home-production rate of married women against distance to the city center. In the baseline, the share of married women in home production rises steeply with distance to the CBD, reflecting both commuting burdens and spatial sorting of household types. Equalizing wages shifts the level of home production down but does not affect the gradient: peripheral areas remain less attached to the labor market because commuting conditions still differ by gender. Equalizing commuting conditions, by contrast, substantially flattens this gradient. The largest participation gains occur in the periphery, where commuting costs were initially most binding. These results highlight that gendered commuting frictions generate spatially unequal access to market work and reinforce specialization within households.

Figure 10 shows welfare changes by household type. Welfare gains are largest for single women in both counterfactuals, with more modest increases for the other groups. Under the counterfactual without discrimination, single men experience only small welfare gains, as they didn't suffer much discrimination in the baseline. Because wages

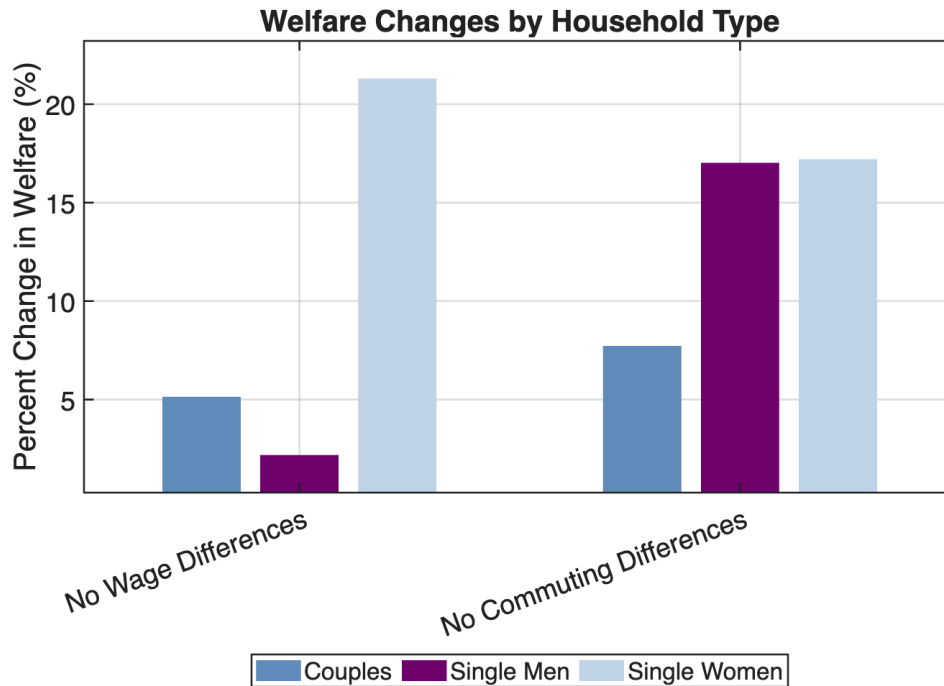
Figure 9: Counterfactual Home Production Shares by Distance to the CBD



Notes: The figure shows the share of married women in home production by neighborhood against distance to the CBD (in kilometers). Each point represents a residential location, with fitted linear trends showing the gradient of home production with respect to distance from the CBD.

rise more for women than for single men, the increase in housing prices partially erode the real income of groups that experience smaller nominal gains. When commuting differences are removed, welfare rises by more than 17% for both single men and single women. The welfare increase for couples is more modest, but larger than when firm discrimination is removed. Equal access to the city's labor market reduces the relative disadvantage of women and peripheral residents, while also lowering commuting disutility for all workers. This shows that improvements in mobility can deliver broad welfare gains even without large changes in wages. The housing market plays a central role in mediating the aggregate effects of these policies. Figure 11 plots residential floor space prices by distance to the CBD. When wage differences are removed, higher wages concentrate demand in central neighborhoods, pushing up rents and steepening the housing-price gradient. The increases in floor space prices absorbs part of the welfare gains from higher wages. In contrast, equalizing commuting conditions flattens the rent gradient: as accessibility improves citywide, households are less constrained to pay a premium for centrality. The resulting equilibrium reallocates both labor and residence

Figure 10: Counterfactual Changes in Welfare by Household Type



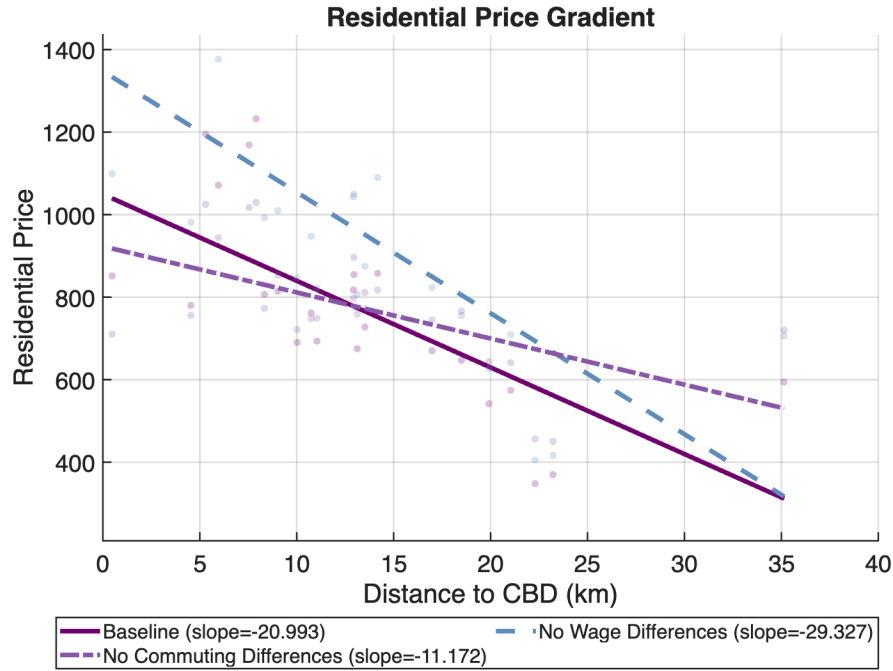
Notes: The bars show the percent changes in welfare relative to the baseline by household type. Blue bars correspond to couples, dark purple bars to single men, and light blue bars to single women.

more efficiently across space, leading to smaller price pressures and higher real welfare as shown in Figure 7.

Both experiments highlight distinct mechanisms linking gendered labor market frictions to the urban spatial equilibrium. Removing wage discrimination raises output by eliminating the taste-based disutility that firms experience when hiring women, which had previously reduced the share of women's production that entered aggregate output. However, the resulting rise in nominal incomes leads to higher housing prices and a mild redistribution of welfare: single women and couples gain, while welfare for single men declines slightly due to higher rents and limited wage adjustments. In contrast, removing commuting differences raises welfare more substantially and more uniformly across groups by improving effective access to jobs, reducing home production, and flattening spatial inequality in participation. Together, the counterfactuals reveal that mobility constraints are central to understanding gender gaps in labor force participation and the geography of home production within cities.

A note of caution is warranted before reading these magnitudes too literally. The

Figure 11: Counterfactual Residential Floor Space Prices



Notes: The figure shows residential floor space prices by distance to the CBD (in kilometers) for baseline and counterfactual scenarios. Each point represents a residential location, with fitted linear trends showing the price gradient.

counterfactual results are likely an upper bound on the actual effects: the model treats individuals as homogeneous within group up to the idiosyncratic Fréchet shock, so every woman is equally responsive to a reduction in commuting costs and every worker equally responsive to a wage increase. In practice, some women face additional constraints, such as childcare obligations, intra-household bargaining, unobserved preferences, that would dampen their response to either intervention. The same homogeneity assumption biases both counterfactuals in the same direction, so although the levels are likely overstated, the comparison between the two remains informative: the larger output and welfare gains from equalizing commuting frictions reflect a genuine difference between the two channels, not an artifact of the homogeneity assumption.

6 Conclusion

This paper studies how commuting frictions shape gender gaps in labor market participation. Using rich survey microdata for São Paulo, I document three facts. First, women—particularly married women—are less likely to work in the market sector when

they live in residential neighborhoods with longer average commutes to the rest of the city. Second, when women work, they commute shorter distances than men but spend at least as much time traveling, because they rely more heavily on slower modes of transport. Third, women drive much less than men even within private-mode users, and this gap is not fully accounted for by household vehicle access or by relative earnings within the couple. To interpret these patterns, I develop a quantitative spatial model in which couples and singles jointly choose where to live and whether and where to work, trading off local amenities, housing costs, and gendered commuting costs. The model reproduces salient features of the data—gender differences in commuting costs and wages directly interacting with location and participation decisions.

Counterfactual exercises show that the gender gap in commuting frictions has large aggregate and welfare consequences. Closing it raises female labor force participation in the city by 10.3 percentage points—an effect larger than closing the gender gap in labor market returns—with the largest gains accruing to married women on the urban periphery. These changes translate into higher aggregate output and welfare, indicating that gender differences in mobility are not only a question of equity but also of efficiency: cities whose transport networks disproportionately disadvantage women forgo a substantial share of their productive potential.

The results carry clear policy implications. Urban transport investments that improve accessibility for women—safer, more reliable, and better-integrated public transport—can meaningfully raise female labor force participation. Policies that expand access to private or flexible modes of transport, or that support dual-earner households in peripheral neighborhoods, are likely to deliver large returns in both welfare and aggregate output.

The paper also opens questions that I do not resolve here: how specific transport infrastructure investments shape gender gaps in commuting outcomes, why women drive less than men even when access and earnings would suggest otherwise. I plan to address some of these in future versions of the paper or in subsequent work.

Two model assumptions are worth flagging. The first is that the mode of transport is taken as exogenous. The descriptive evidence in this paper makes clear that mode choice is itself a central margin of the gender gap: women drive much less than men even at parity in income, household structure, and vehicle access. Endogenizing mode choice is therefore not merely a technical refinement but a way to study the mechanism

behind one of the paper’s central facts, and it would require a deeper understanding of why women are less likely to drive—norms, safety, licensing, preferences—and which policies can equalize access to fast modes when driving is itself constrained.

The second assumption is selection into home production. In the model, location contributes to sorting into home production but agents are otherwise homogeneous up to the idiosyncratic Fréchet shock. This makes the participation decision rely heavily on commuting costs, when in practice some women will be much less responsive to commuting cost changes than others. Allowing for unobserved heterogeneity, and for selection on observables such as skill, age, or household composition, would sharpen the welfare counterfactuals. Doing so is harder than it appears: once workers differ in ways that affect both participation and location, equilibrium requires tracking the full joint distribution of types across space, which is itself endogenous to wages, rents, and amenities. Adding selection on observables compounds the difficulty, since identifying the participation margin separately from the location choice requires variation in observable characteristics that shifts the outside option without affecting sorting conditional on working—an exclusion restriction that is difficult to find in practice, since the same characteristics that determine whether someone works also tend to determine where they work and live.

The avenues for future research these questions open are promising. They come with real challenges—tracking heterogeneous types in equilibrium, identifying the participation margin separately from location choice, observing intra-household bargaining over mode—but the questions themselves are interesting and consequential.

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A Data and Context

In this Appendix Section, I first provide some additional context about the city and where it stands relative to other large cities in the world. Then, I describe with detail the datasets used in the paper, and the creation of the relevant variables, providing additional description of the sample. I follow by discussing in detail how I created the travel time matrices used in the model estimation.

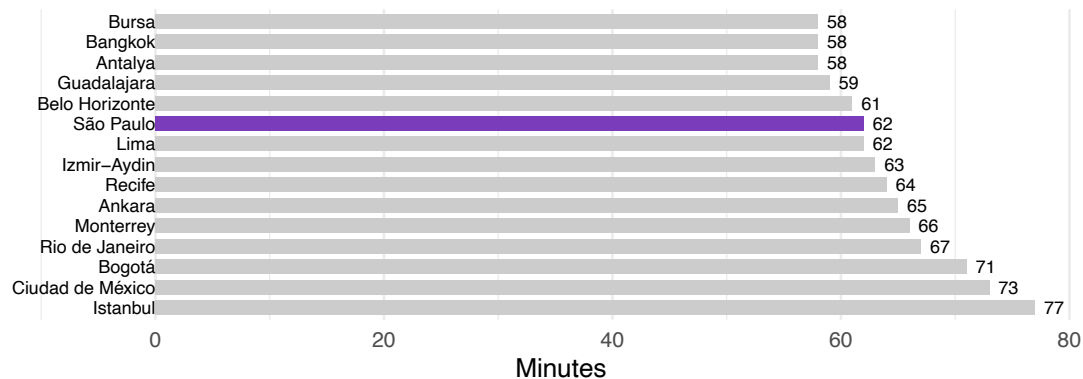
A.1 Additional Context

São Paulo is the largest city in Latin America, and the city alone accounts for over 10% of Brazil’s GDP. It also shares many features of other global megacities, with dense urban cores surrounded by sprawling peripheries, long commutes, and stark spatial inequalities.

Figure A1: Cities with Longest Commuting Time in the World

Average Commute Time by Public Transit (one way), 2022

Data: Moovit – includes walking, waiting, and in vehicle time



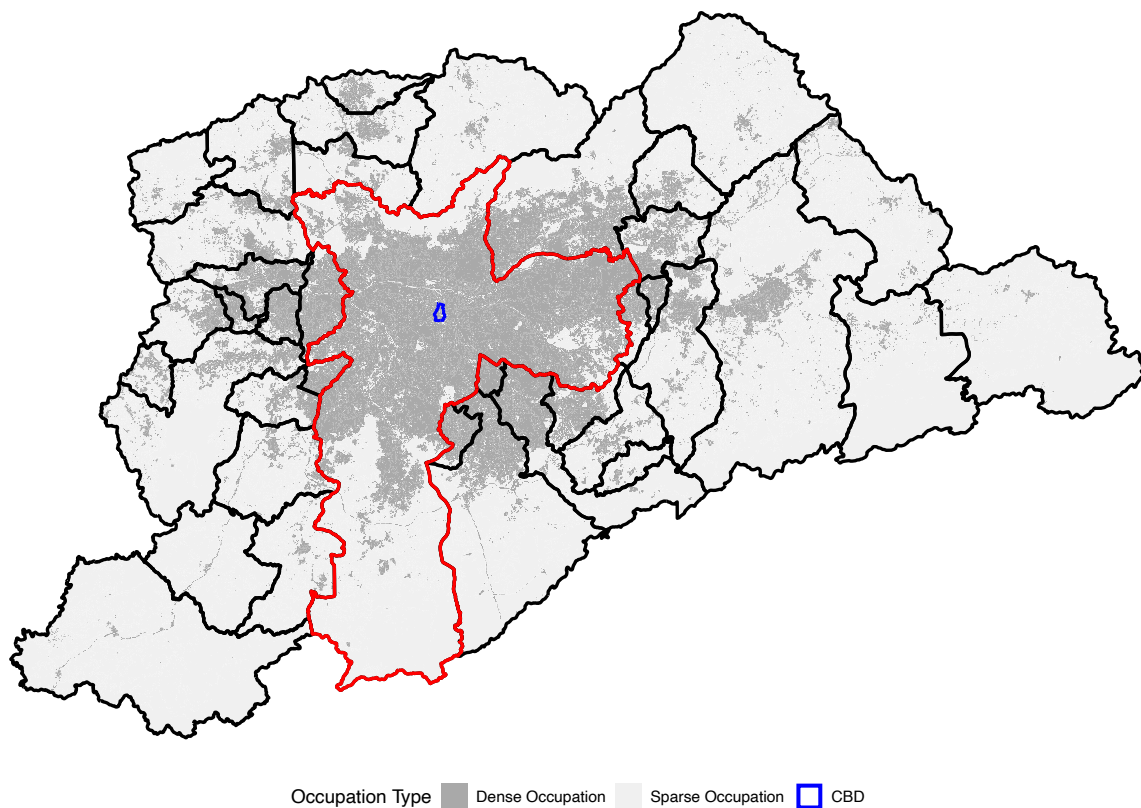
Notes: The figure displays the cities with the highest reported average commute times among large metropolitan areas worldwide. Average one-way commute times are reported by Moovit for 2022 and include walking, waiting, transfers, and in-vehicle time on public transit.

São Paulo city is part of the São Paulo Metropolitan Region, which comprises 39 municipalities in total. The region has a population of 19.6 million inhabitants, more than half (57.2 percent) of whom live in São Paulo municipality alone. It is the fourth most populous metropolitan region in the world. Although the city covers only 19.2 percent of the region’s area, it concentrates 56.2 percent of its population. Nearly 85 percent of those who work in the city also reside there. Figure A2 highlights São Paulo within the metropolitan region and shows the distribution of Local Climate Zones, distinguishing

areas of dense and sparse urban occupation.²⁸ Most of the densely built-up areas in the region are located within the municipality of São Paulo.

I focus on the city of São Paulo rather than the entire metropolitan region due to data limitations. Data on commercial floor space prices are unavailable for the other municipalities in the region. In addition, public transport networks are largely managed at the municipal level, which makes consistent network data across the region difficult to obtain. This is particularly binding for the model estimation, which depends on these additional variables. The stylized facts presented in Subsection 2.3 are robust to considering all the metropolitan region, rather than just the city.

Figure A2: São Paulo Metropolitan Region and Urban Occupation Type



Notes: The map shows the 39 municipalities of the São Paulo metropolitan region and São Paulo municipality is highlighted in red. The data contains 100m2 satellite images classified into Local Climate Zones (LCZ). Dark grey considers any urban occupation, from compact high-rise (DN 1) until large lowrise (DN 8). The light grey area accounts for water, vegetation and sparsely built area (DN 9). Data and shapefile are from [Ferreira \(2019\)](#). Highlighted in blue is the center business district (CBD).

²⁸[Ferreira \(2019\)](#) classify satellite imagery according to the Local Climate Zone scheme.

A.2 Origin-Destination Surveys

The main data source used in this paper are the Origin-Destination Surveys for São Paulo. The *Pesquisa Origem e Destino* is a large-scale household survey conducted by the Department of Metropolitan Transportation of São Paulo State. It collects information on residents' daily travel patterns, socioeconomic characteristics, and mobility behavior. The 2007 wave covered around 30 thousand randomly selected households and 120 thousand individuals, distributed across 460 survey zones that span the entire São Paulo Metropolitan Region.

The survey provides micro data structured at the household, individual, and trip levels. The household questionnaire records information on dwelling characteristics, tenure, and ownership of durable goods and vehicles. At the individual level, it includes demographics, education, employment status, occupation, income, and schooling or workplace locations. The trip module reports all journeys made on the day prior to the interview, detailing origin and destination zones, purpose, departure and arrival times, modes of transport (up to four per trip), walking segments, and travel costs. This structure enables the construction of complete home-work commutes and the analysis of mode choice, travel time, and labor market accessibility.

Household type. To identify couples, I use the household position variable reported in the survey. I classify as couples the household head and their reported spouse or partner, provided that both individuals are cohabiting in the same household. Each person is coded according to their relationship to the household head, which can be spouse or partner, child or stepchild, other relative, non-relative household member, live-in employee, or relative of an employee. Because this classification is defined relative to the household head, I am unable to detect additional couples within the same household when neither partner is the head. As a result, my measure may undercount cohabiting couples in extended or multi-family households. Close to 60 percent of individuals in the sample live in nuclear households (headed by a couple), while around 20 percent reside in non-nuclear or multi-generational families that include additional adults beyond parents and children. These cases range from siblings or friends sharing a dwelling to extended families cohabiting in the same household. The remaining individuals live in single adult households. All adults not classified as part of a couple are considered singles in the analysis, independent of the type of household they live in. An adult single child residing with parents is considered a single person in the sample and model.

Wealth. I construct a measure of household wealth quintiles based on asset ownership, following the methodology commonly used in the Demographic and Health Surveys (Smits and Steendijk, 2015). The measure relies on information about durable goods and household amenities, under the assumption that variation in ownership and access to these assets captures long-run differences in economic status. Specifically, I include the number of bathrooms and domestic employees, as well as ownership of items such as cars, washing machines, refrigerators (one- or two-door), freezers, microwaves, dishwashers, DVD players, motorbikes, and bicycles. I apply Principal Component Analysis (PCA) to these asset variables and use the first principal component as a summary wealth index. Households are then ranked by their PCA score and divided into five equally sized groups (quintiles) using sampling weights. These measures are highly correlated with income and education level of the household head.

Sample. To define the sample for the analysis I use a variable called activity status, which is tabulated by gender in Table A1. I remove from the analysis students and retirees, who account for less than 6% of the adult sample aged between 18 to 55, and keep only individuals who are working or not working and house wives.

Table A1: Activity status by gender

Activity Status	Male (%)	Female (%)
Worker	84.17	63.86
House wife	0.18	17.07
Not working	10.38	13.38
Retired	2.63	3.2
Student	2.37	2.48
N	16638	19481

Source: Household Travel Survey of 2007. The sample contains individuals aged 18 to 55, residing in the municipality of São Paulo. Working also includes occasional workers, and any type of job or employment.

Home Production. I pool together those who are not working and those classified as housewives under the category of home production. Out of the remaining sample, approximately 23% are out of the labor market—an important share, which motivates my focus on the extensive margin of unpaid economic activity. Although the survey

provides detailed trip information that can be used to infer how people allocate their time, more than 60% of individuals outside the labor market reported making no trips on the previous day, limiting what can be inferred about their activities. For those not working, the data do not indicate whether they are actively seeking employment. However, I can identify trips made for job search, and only about 2% of individuals in home production reported such trips, consistent with evidence from recessions showing that once individuals exit the labor market, they largely substitute time in market work for home production and leisure (Aguilar et al., 2013).

Furthermore, most of the differences by gender in activity type choices occur in the extensive margin. Among those in market work, they may be working in either regular jobs or flexible jobs. Table A2 summarizes key characteristics by work status. The male–female ratios indicate that men slightly outnumber women among those in regular and flexible jobs (around 54–53% men, respectively), but women account for roughly three quarters of individuals in home production. Workers in flexible jobs are more likely to work from home (23% versus 2% among regular workers) and have shorter commutes, though they also earn lower average incomes. Individuals in home production make fewer daily trips overall, but a larger share of those trips are related to care (consumption, health or education related), reflecting the concentration of unpaid domestic responsibilities among women.

Table A2: Activity Status and Characteristics

	Regular job	Flexible job	Home Production
Male-Female ratio	1.18	1.13	0.31
Share working from home	0.02	0.23	–
Average income (R\$)	1,392.97	1,090.61	–
Average commute (minutes)	60.99	41.64	–
Number of trips	2.43	2.25	1.17
Number of care trips	0.46	0.55	0.81
<i>N</i>	17,247	9,657	6,420

Notes: Sample includes adults aged 18 to 55 that are either working or in home production. Regular jobs include formal workers, civil servants, business owners, practitioners. Flexible jobs include informal workers, self-employed, family business owner or employees. Home Production includes individuals not working and house wives. Income and commute variables are conditional on being employed. Trips refer to all reported trips on the previous day; care trips include education, health and consumption purposes.

Reported Commuting Time. To compute reported commuting time between the residence and workplace, I consider direct trips between these two locations or sequences of trips separated by an intermediate stop of less than ten minutes. In the latter case, I sum the durations of the consecutive trips to obtain the total commuting time. Such chained commutes represent less than 0.5% of all commutes in the sample.

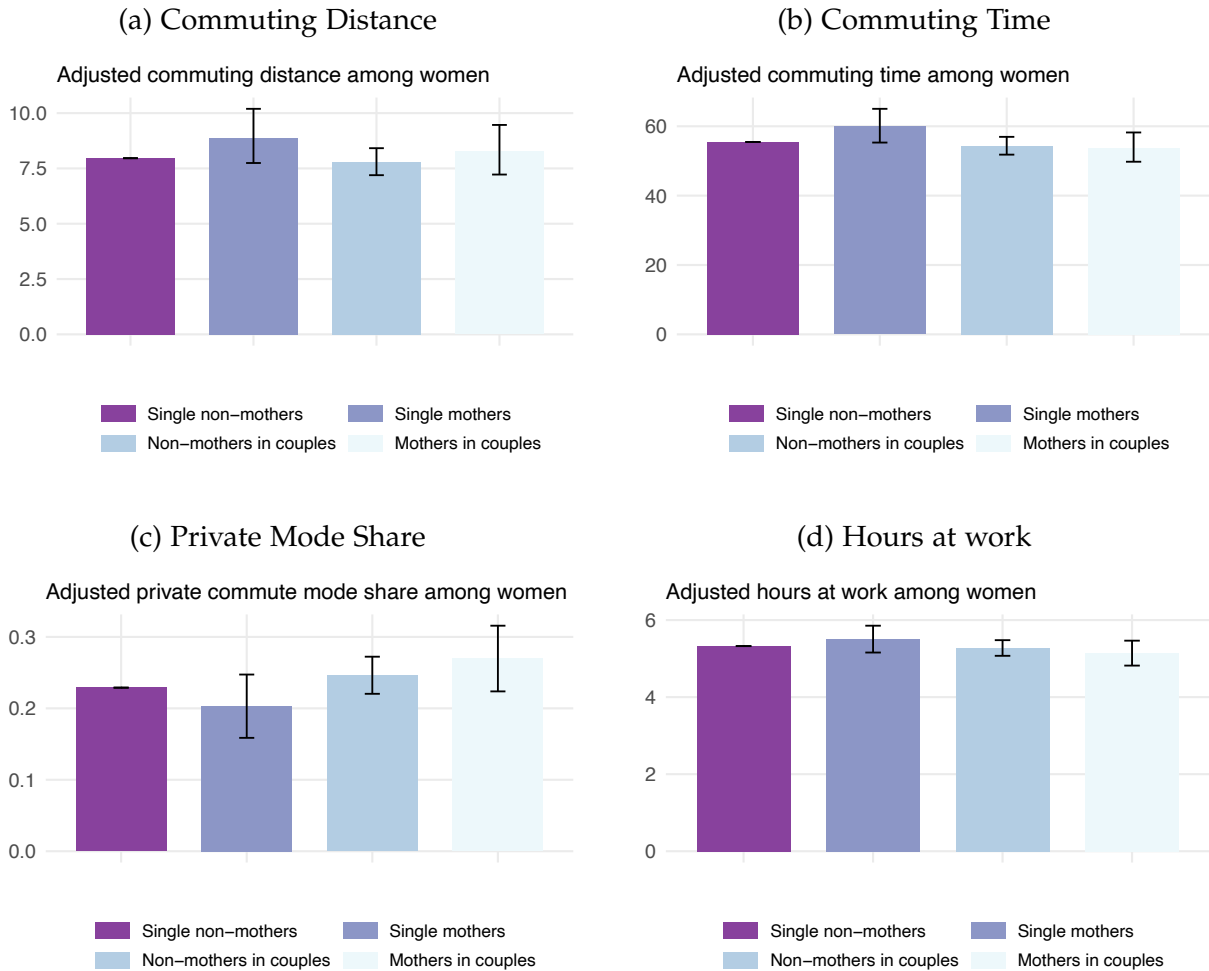
Children. A natural concern about the gender gap in labor market participation is that it is driven by children: women’s labor supply drops sharply after the birth of a first child while men’s is largely unaffected, a pattern documented in many countries (Kleven et al., 2019, 2024). In my model, I treat the relevant division as couple versus single rather than mother versus non-mother, for two reasons. First, couple status and motherhood are highly correlated in my sample: only 10.9% of individuals are singles with children, against 35.5% who are singles without, while individuals in couples are split roughly evenly between those with children (26.4%) and those without (27.1%).

Second, when I compare commuting and time-use outcomes across the four-way breakdown by motherhood and couple status, single mothers tend to be the outlier; differences across the other three groups are generally small and statistically insignificant.

I replicate Figure 3 for women only, splitting by motherhood and couple status rather than by gender and marital status. Figure A3 reports the adjusted commuting outcomes. Commuting distance (Figure A3a) is statistically indistinguishable across the four groups. Commuting time (Figure A3d) is similar for most women, but single mothers commute on average about five minutes longer than the others. Single mothers are also significantly less likely to commute by private mode than every other group of women. The single-versus-married contrast is sharper among mothers than among non-mothers, and the mode gap most naturally reflects access to a household vehicle: married women can typically draw on a partner’s car, single mothers usually cannot.

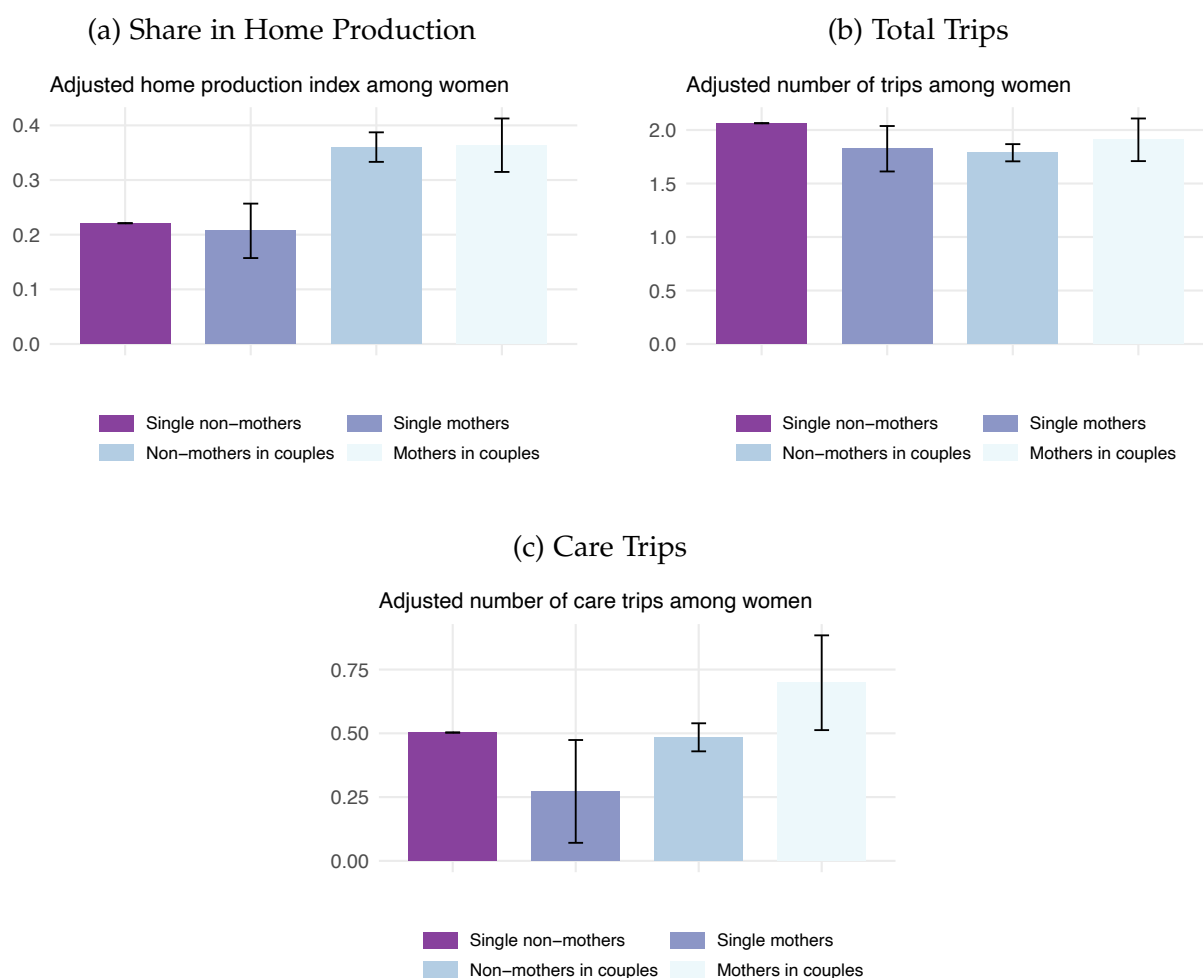
Figure A4 points out that when it comes to being in home production, the key distinction is being in a couple rather than being a mother. On time-use outcomes and the likelihood of being in home production, single mothers again stand out: they work more hours and make fewer total trips than the other three groups, which themselves report similar numbers of daily trips and time at work. Married women with children do make more care trips—trips for education, shopping, or health (Figure A4c)—but this does not come at the cost of hours at work or other commuting outcomes.

Figure A3: Adjusted Commuting Outcomes by Couple Status and Motherhood (Women)



Notes: The figure reports adjusted commuting outcomes for women in the regression sample, by marital status and presence of children. Adjusted means come from a regression on indicators for being in a couple, being a mother, and their interaction, controlling for age and age squared, the number of children aged 0–3 and 4–10, and the number of elderly aged 75+, with residence-district, workplace-district, education, and wealth-quintile fixed effects.

Figure A4: Adjusted Time-Use and Labor-Supply Outcomes by Couple Status and Motherhood (Women)



Notes: The figure reports adjusted outcomes for women in the regression sample, by marital status and presence of children. Care trips are trips reported for education, health, or consumption purposes; total trips include all reported trips on the previous day; hours at work refers to hours worked on the previous day. Adjusted means come from a regression on indicators for being in a couple, being a mother, and their interaction, controlling for age and age squared, the number of children aged 0–3 and 4–10, and the number of elderly aged 75+, with residence and workplace location, education, and wealth-quintile fixed effects.

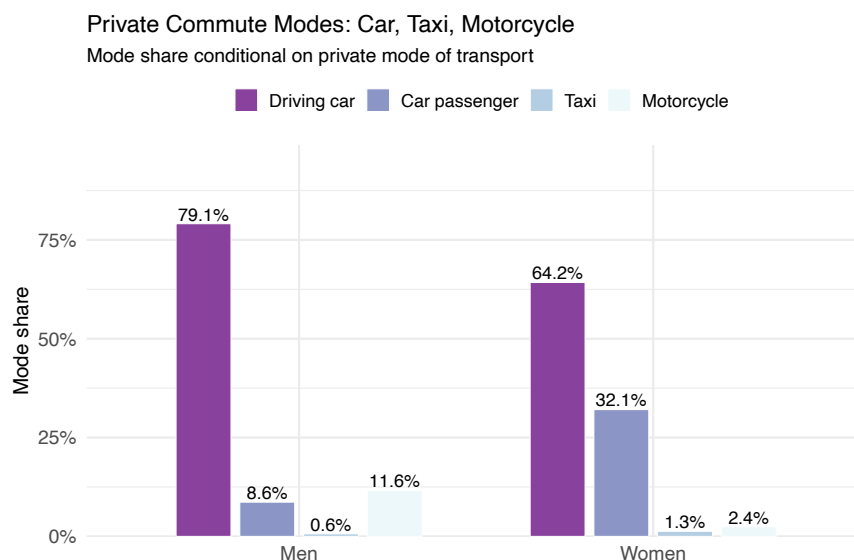
A.3 Why Women Don't Drive

A central pattern in the commuting data is that women are much less likely to drive and use private modes of transport than men. This subsection documents the size of the gap and shows that it is not fully accounted for by the most natural confounders: access to a household vehicle and the distribution of earnings within the household. While I cannot pinpoint a single mechanism behind the residual gap, I aim to describe the patterns and show that they are large enough and persistent enough across slices of the data that they justify treating the cost of private transport as asymmetric by gender.

Within private-mode users, women drive much less than men. I begin by isolating workers who report commuting by a private mode at all (car, taxi, or motorcycle) and comparing how they use it. Figure A5 shows that, conditional on commuting by private mode, men overwhelmingly drive themselves: 79.1% of male private-mode commuters report driving a car, compared with 64.2% of women. Women are correspondingly much more likely to be car *passengers*—32.1% versus 8.6% for men—a nearly four-fold gap. Motorcycles are also a much larger share of private-mode commuting for men than for women. The gender gap in commuting therefore is not only that women use private transport less; even within the set of workers who do, women are substantially less likely to be the person behind the wheel.

Vehicle access does not fully explain the gap. A first natural explanation is that women drive less because they have less access to a household vehicle. Figure A6a plots the distribution of household vehicle ownership in the analysis sample: 38.1% of individuals live in households with no vehicle, 42.5% in households with exactly one, and 19.4% in households with two or more. Vehicle access is therefore a real constraint for a non-trivial share of the sample, but a clear majority of workers live in households with at least one vehicle. Figure A6b then plots the share commuting by private mode separately for men and women, conditional on household vehicle ownership. The gender gap is largest in single-vehicle households, where men are more than twice as likely as women to commute by private mode (52.2% versus 25.1%)—consistent with intra-household allocation of the single car favoring men. Importantly, the gap does not disappear in households with two or more vehicles: men's private-mode share rises to 82.9%, but women's only to 69.5%, leaving a gap of more than 13 percentage points even when at least one vehicle is plausibly available to each spouse. Vehicle access therefore

Figure A5: Drive vs. Be Driven: Mode Shares Among Private-Mode Commuters

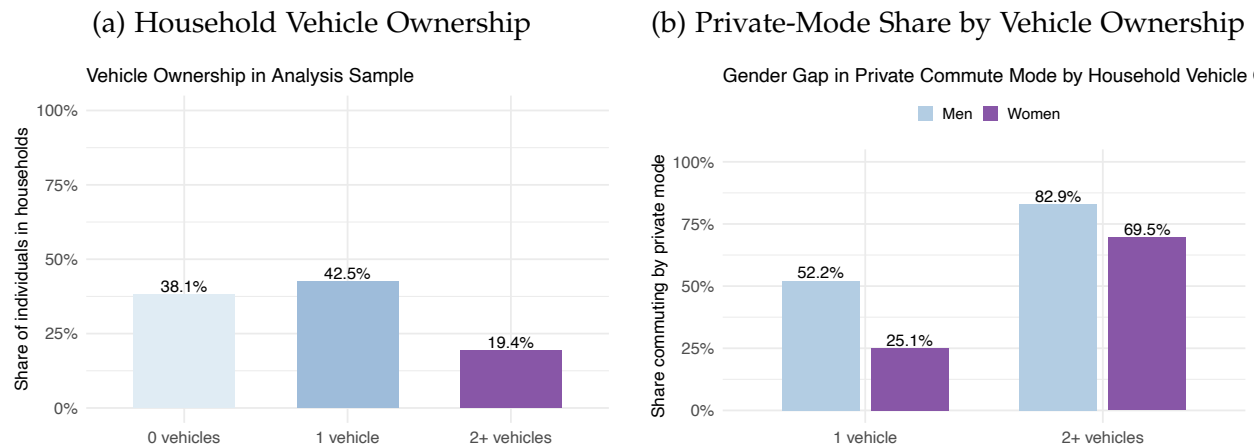


Notes: The figure reports mode shares among workers who commute by a private mode of transport (car, taxi, or motorcycle), separately for men and women, in the 2007 Origin–Destination survey. “Driving car” indicates the respondent reports driving themselves; “car passenger” indicates being driven by someone else. Conditional on using a private mode, men are substantially more likely than women to be the driver, while women are much more likely to be passengers.

matters, but it is not the whole story.

Earnings and observables do not fully explain it either. A second natural explanation is intra-household bargaining: if husbands typically earn more, they may have first claim on the household car. Figure A7a plots the share of men and women in couples commuting by private mode, broken down by relative spousal income. When the wife earns less, the gap is large (38.1% for husbands versus 17.5% for wives). When the wife earns about the same, both shares rise and the gap narrows (43.0% versus 34.5%). When the wife earns more, the husband’s share *falls* to 31.3% and the wife’s rises slightly to 27.4%, but men still drive more than women even in households where women out-earn their partners. The gradient is consistent with a bargaining channel, but it does not close the gap. Figure A7b extends this to the full sample of men and women (including singles) by reporting adjusted private-mode shares from a regression that conditions on income, education, residence location, household structure, and other observables. After this adjustment the raw difference compresses substantially, but a gap of about six percentage points remains: the predicted private-mode share for men is 41.8%, compared with 35.7% for women. In other words, even comparing observationally equivalent men

Figure A6: Vehicle Ownership and the Gender Gap in Private-Mode Commuting



Notes: Panel (a) reports the distribution of household vehicle ownership across individuals in the analysis sample (2007 Origin–Destination survey). Panel (b) reports the share of workers commuting by private mode (car, taxi, or motorcycle), separately by gender and by household vehicle ownership. The gender gap is largest in single-vehicle households and narrows in households with two or more vehicles, but a substantial gap remains even when vehicle access is unlikely to be the binding constraint.

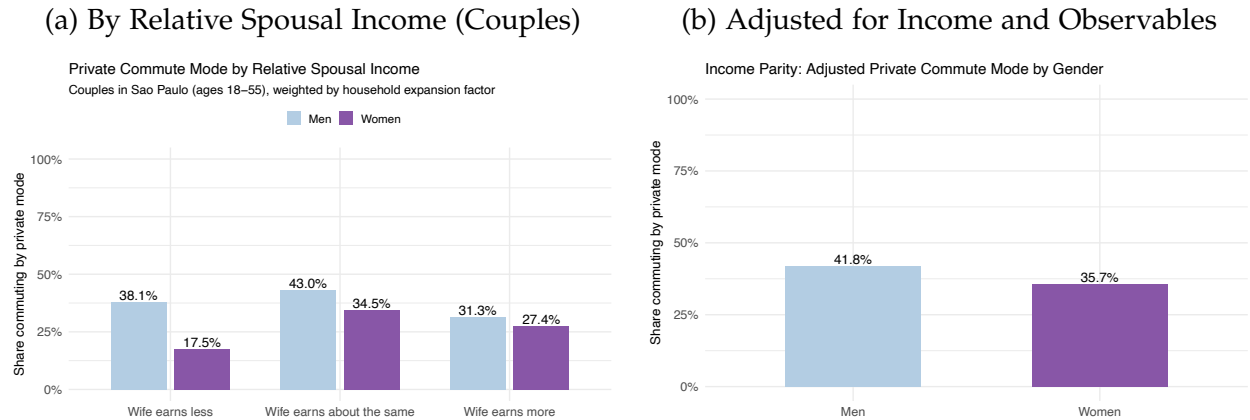
and women—same income, same residence, same household composition—women are notably less likely to commute by private transport.

Taken together, these patterns establish three points. First, women use private modes of transport substantially less than men, and they drive even less frequently. Second, this gap is not simply a story about vehicle access: it persists in households with multiple vehicles. Third, it is not simply a story about earnings or bargaining either: it persists when wives out-earn their husbands and after conditioning on income and a rich set of individual and household observables. I cannot adjudicate among the remaining candidate explanations—gender norms over who drives in the household, safety concerns linked to parking and late-night travel, differences in licensing rates, or unobserved preferences—using these data alone. What I can establish is that the gender gap in driving is large and persistent.

A.4 Travel time matrix

I compute the travel time matrices between neighborhoods by first identifying the population-weighted centroid of each neighborhood. Using a multimodal transport network router ([Pereira et al., 2021](#)), I calculate travel times between these centroids based on network data from OpenStreetMap and General Transit Feed Specification (GTFS) files for São Paulo. The network includes all major public transport modes—underground

Figure A7: Income, Bargaining, and the Gender Gap in Private-Mode Commuting



Notes: Panel (a) reports the share of husbands and wives in couples (ages 18–55, weighted by household expansion factor) who commute by private mode, broken down by relative spousal income (wife earns less, about the same, or more than the husband). Panel (b) reports private-mode shares for men and women predicted from a regression that conditions on individual income, education, residence and workplace location, household structure, and other observables. Both panels show that the gender gap narrows considerably when income and bargaining position are accounted for, but does not fully close.

and overground rail, as well as municipal and inter-municipal buses—managed by two separate transport authorities.

In São Paulo, two agencies are responsible for maintaining GTFS data. *SPTrans*, the municipal transport management company, provides route information for buses and rail services operating within the city. The earliest available data from *SPTrans* are from January 2012. I adapt this network to represent conditions in 2007 by removing rail lines and stations built after that year and assuming limited changes to the bus network. The inter-municipal bus network, managed by the Metropolitan Urban Transport Department (EMTU), represents a smaller share of total transit. As the earliest EMTU data are from 2017, I use the same network to compute travel times for both 2007 and 2017.

The routing algorithm returns a distribution of travel times between each pair of centroids, reflecting variation in possible routes, departure times, and service frequencies. I calculate travel times for trips departing on a Friday at 6 a.m. and use the median travel time between each pair of points. Separate matrices are computed for public transport, private car, and walking. To obtain effective travel times by group, I weight these mode-specific matrices using observed mode shares by gender and marital status. Table A3 reports these shares, showing that women—especially married women—use public transport more intensively than men, while men rely more on private transport. Walk-

ing represents a small but non-negligible share across all groups, particularly among women. Public transport is the predominant commuting mode across all groups, indicating that most of the city’s workers rely on it as their main means of travel to work.

Table A3: Transport Mode Shares by Group for Commuting Trips

	Public Transport	Private Transport	Walking
Married Women	0.75	0.21	0.04
Married Men	0.58	0.40	0.02
Single Women	0.79	0.19	0.02
Single Men	0.64	0.33	0.03

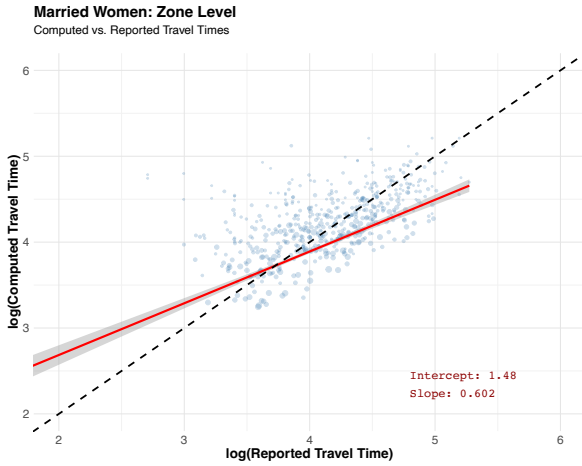
Note: Mode shares are computed from the 2007 Origin–Destination survey for employed individuals aged 18–55 living in the city of São Paulo. Private transport includes cars, taxis, and motorcycles. The shares are computed for those who do not commute within their own neighborhood, as they are much more likely to walk.

Validation. To validate the median travel times, I use the reported trip duration between neighborhoods from the Origin–Destination survey, and I plot them against the computed travel times using the routing algorithm. The comparison is in logs, and observations are weighted by the number of reported trips on each bilateral, since some origin–destination pairs are observed only a handful of times and a level-on-level comparison is overly sensitive to the resulting noise as well as to a few extreme long commutes. Working in logs also has the natural interpretation of comparing relative rather than absolute differences and absorbs multiplicative measurement error—a +20-minute discrepancy is not the same kind of error at 20 minutes as at 120 minutes.

Figure A8 reports the comparison separately for each gender–marital status group. The relationship between log reported and log computed travel times is positive and tightly clustered for all four groups, but the slopes are below one—ranging from roughly 0.60 to 0.67—rather than tracking the 45-degree line exactly. The key feature of the figure is that the slopes and intercepts are very close across married women, married men, single women, and single men. While the routing algorithm therefore does not reproduce reported trip durations one-for-one, the bias it introduces is symmetric across groups, so the computed matrices preserve the relative ranking of origin–destination pairs across gender and marital status. Any group-specific differences in commuting costs identified using these matrices are unlikely to be driven by measurement error in the travel times themselves.

Figure A8: Validation of Computed Travel Times by Group

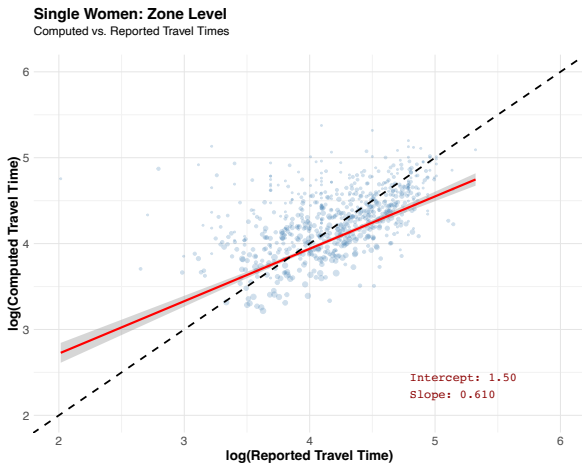
(a) Married Women



(b) Married Men



(c) Single Women



(d) Single Men



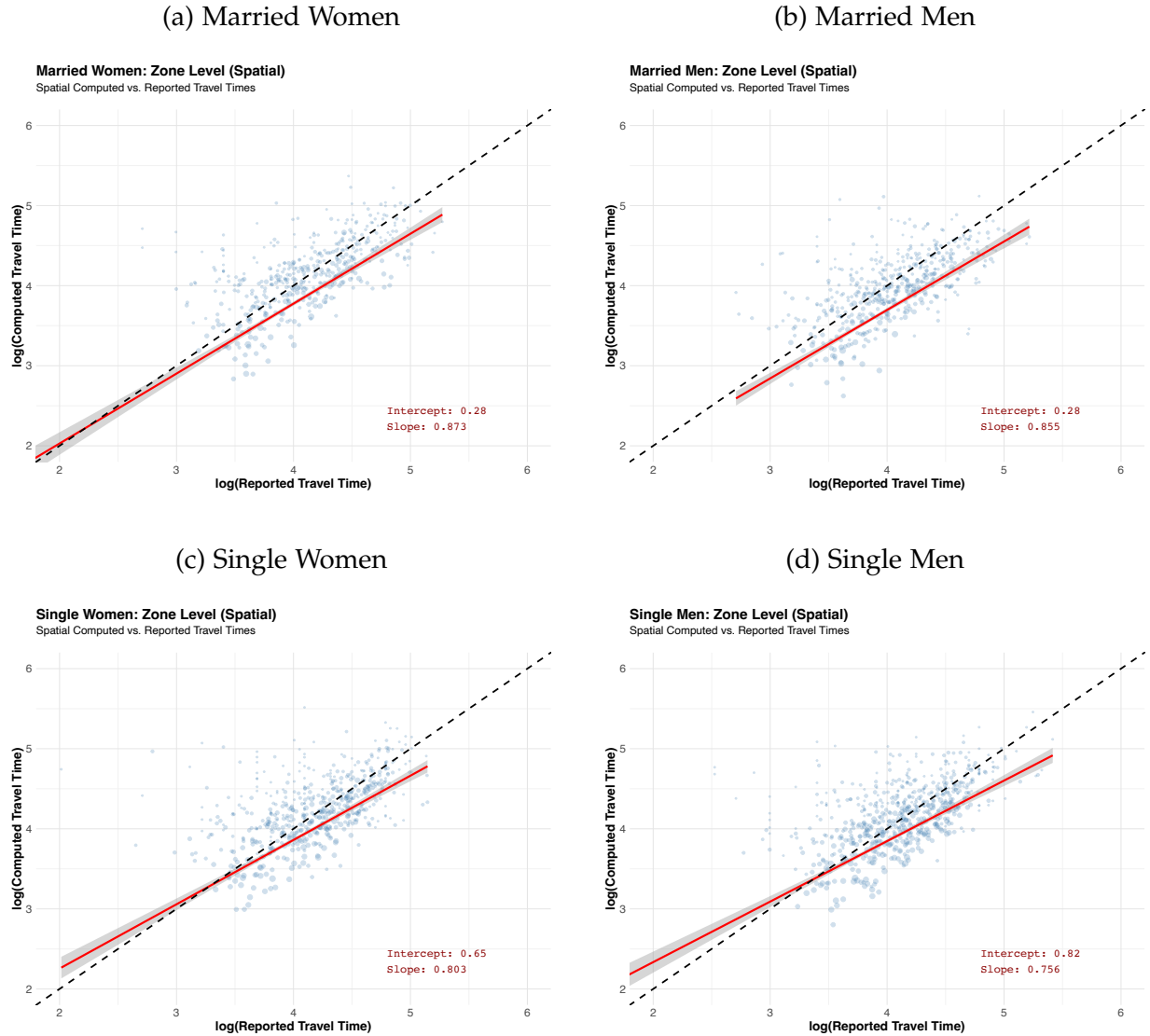
Notes: Each panel plots log reported commuting travel times from the 2007 Origin–Destination survey (vertical axis) against log computed travel times from the multimodal routing algorithm (horizontal axis), at the origin–destination zone level. Observations are weighted by the number of reported trips on each bilateral. Computed travel times are weighted by group-specific city-level mode shares from Table A3. Panel (a) shows married women, Panel (b) married men, Panel (c) single women, and Panel (d) single men. The dashed line is the 45-degree line. The slopes and intercepts are similar across all four groups, indicating that any measurement error in the computed travel time matrices is roughly symmetric across gender and marital status.

There are two natural reasons why the computed matrices fail to reproduce reported travel times exactly. First, the routing algorithm uses the OpenStreetMap road network, which encodes free-flow speeds based on road type rather than realistic traffic conditions. In a mega-city like São Paulo, congestion is severe and highly heterogeneous across the day and across corridors, so private-transport travel times computed on the OSM network are best interpreted as a lower bound on actual driving times—particularly on long, congested radial corridors where the gap between free-flow and rush-hour speeds is largest. Because congestion delays do not enter the OSM-based computation but do affect the times people report, they shift reported times above computed times in a way that varies across bilaterals and depresses the slope of reported on computed travel times below one. Second, the baseline matrix weights mode-specific travel times by mode shares computed at the city level, after excluding own-zone (intra-zone) flows. City-level shares are a useful summary of how each group typically commutes, but they may misrepresent the mode mix on specific origin–destination pairs: bilaterals that are poorly served by public infrastructure tend to attract a disproportionate share of car commuters, while bilaterals well-served by rail or BRT corridors tend to attract more public-transport commuters. Using a single set of city-level shares smooths over this heterogeneity, and may bias travel times upward on car-dominated corridors and downward on transit-dominated corridors.

To assess the importance of this second channel, I also compute an alternative travel time matrix in which the mode-specific matrices are weighted using *neighborhood-level* mode shares—that is, the share of each mode on each origin–destination pair, computed group-by-group from the Origin–Destination survey. Figure A9 replicates the validation exercise using these neighborhood-specific weights. The slopes increase substantially across all four groups, rising from the 0.60–0.67 range in the baseline to a 0.75–0.87 range under neighborhood-level weighting, and the cross-group symmetry of the bias is preserved. The size of this improvement indicates that a meaningful share of the gap between reported and computed travel times in the baseline matrix is attributable to the use of city-level mode shares, rather than to the OSM network alone. The residual gap between the slope and one in the spatial version—which is what cannot be closed by refining the mode-share weighting—is consistent with the absence of congestion in the OSM network and with reporting noise in self-reported trip durations. Either way, the qualitative comparisons across groups are not sensitive to which weighting scheme is

used.

Figure A9: Validation of Computed Travel Times Using Neighborhood-Level Mode Shares



Notes: The figure replicates Figure A8 using mode shares computed at the origin–destination neighborhood pair level rather than at the city level. For each gender–marital status group, the mode-specific travel time matrices are weighted by the share of each mode observed on each bilateral in the 2007 Origin–Destination survey. Both axes are in logs, and observations are weighted by the number of reported trips on each bilateral. Panel (a) shows married women, Panel (b) married men, Panel (c) single women, and Panel (d) single men. The dashed line is the 45-degree line. The slopes are markedly closer to one than in Figure A8 and the cross-group symmetry of the residual bias is preserved.

A.5 Additional Tables

Table A4: Probability of Home Production by Log Avg. Commute Time

	P(Home Production)			
	Singles (1)	Couples (2)	Singles (3)	Couples (4)
Log Avg. Commute Time	0.12*** (0.03)	0.05*** (0.02)	0.03 (0.03)	-0.05*** (0.02)
Log Avg. Commute Time $\times$ Female	0.09** (0.04)	0.13*** (0.04)	0.08** (0.04)	0.14*** (0.04)
Observations	16,232	16,541	16,232	16,541
Individual-level controls			✓	✓
Household-level controls			✓	✓
Education FEs			✓	✓
Wealth quintile FEs			✓	✓

Notes: Robust standard errors in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: SP origin destination survey for 2007. Sample includes adults aged 18 to 55 living in the city of São Paulo who are working or in home production. Controls include age, age squared, household size, number of 0-3 year-olds, 4-10 year olds and number of 75+ in the household in the household. Fixed effects for education level and for wealth quintile based on household assets are included. Average commute time at the residence district level includes all reported commuting times.

Table A5: Commuting Outcomes by Gender and Marital Status

	Dependent Variable		
	Log Commuting Distance	Log Commuting Time	Share Private Transport
	(1)	(2)	(3)
Female	-0.04 (0.04)	0.10*** (0.03)	-0.13*** (0.01)
Married	0.14*** (0.04)	0.06* (0.03)	0.05*** (0.01)
Female $\times$ Married	-0.26*** (0.06)	-0.15*** (0.04)	-0.03 (0.02)
Observations	19,325	19,325	19,325
Residence Bin FE	✓	✓	✓
Education FE	✓	✓	✓
Wealth quintile FE	✓	✓	✓

Notes: Each column reports coefficients from a regression of the outcome on gender and marital status indicators and their interaction. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Controls include age, age squared, household size, number of 0-3 year-olds, 4-10 year olds and number of 75+ in the household in the household. Fixed effects for education level, for wealth quintile, and for location bin are included. Location bins are based on the distance to the CBD, and each bin includes 25% of the sample.

Table A6: Different Returns in the Labor Market by Gender and Marital Status

	Log(Earnings) (1)
Married Woman	-0.31*** (0.02)
Single Woman	-0.32*** (0.02)
Married Man	-0.11*** (0.02)
Age	0.05*** (0.005)
Age ²	-0.00*** (0.00)
Log Time at Work	0.03* (0.02)
Observations	12,043
R ²	0.49
Sector fixed effects	✓
Occupation fixed effects	✓
Job Type fixed effects	✓
Education fixed effects	✓

Notes: The dependent variable is the log of the individual's wage. Controls include age, job characteristics, and education. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.6 Commuting outcomes

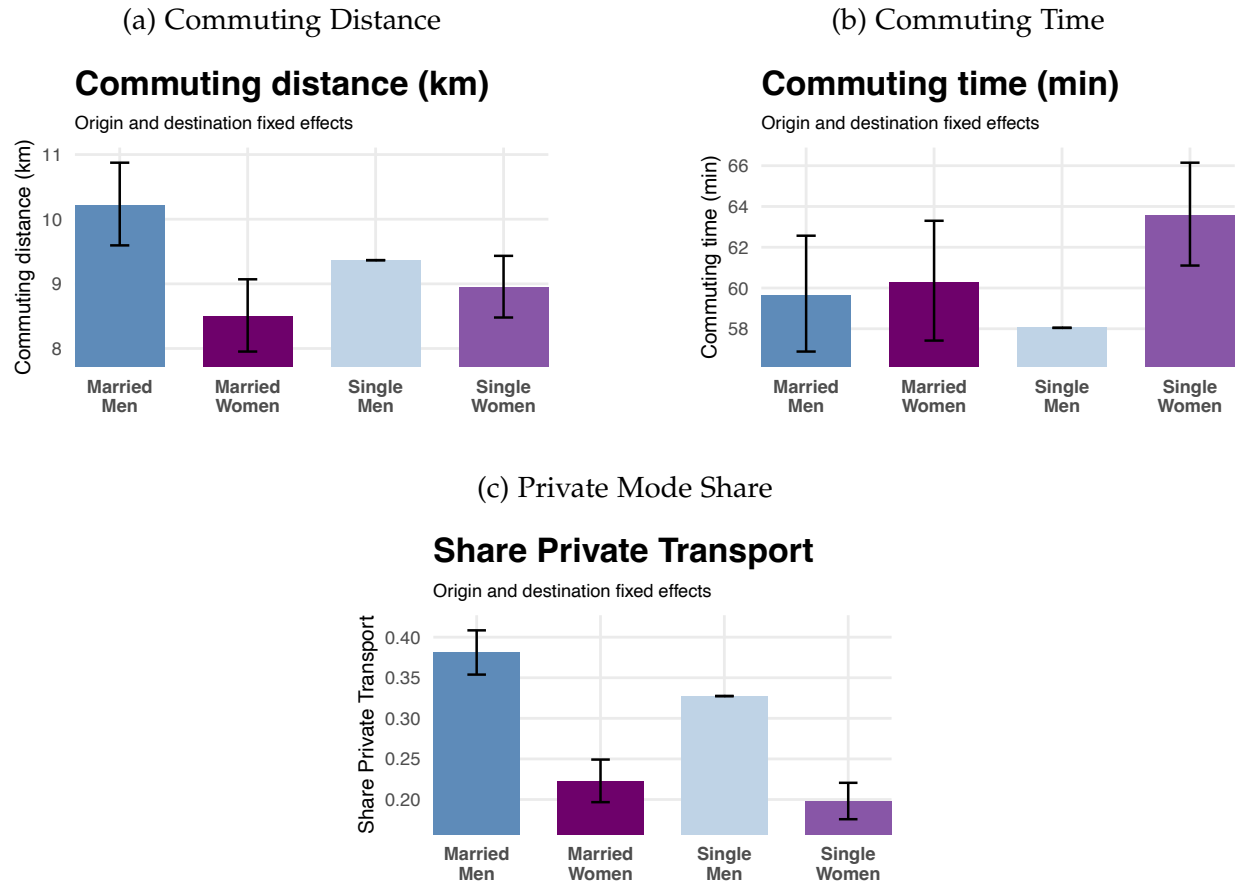
This subsection presents three complementary cuts of the commuting-outcome regressions reported in Figure 3 that probe whether the gender–marital gaps in commuting distance, time, and private-mode share are explained by other factors.

Sorting across residence and workplace locations. A natural concern with the baseline gaps is that they may partly reflect differential sorting across the city—women may systematically live closer to the center or work in different parts of the city than men, so that apparent group-level differences in distance, time, and mode share could capture the geography of where people live and work rather than differences in commuting behavior. Figure A10 re-estimates the same group-level differences after additionally absorbing residence and workplace fixed effects, comparing only individuals who live in the same residence zone and work in the same workplace zone. The estimated gaps are essentially unchanged relative to the baseline: married women still commute substantially shorter distances than married men, the time gaps remain small but in the same direction, and the private-mode gap between men and women is essentially identical to the unconditional version. Differential sorting across residence and workplace locations is therefore not the main driver of the observed differences.

Conditioning on public transport use. A second cut isolates the role of mode choice. The mode-share gap in Figure 3c suggests that some of the distance and time differences across groups operate through who uses public transport in the first place. Figure A11 restricts the sample to workers commuting by public transport and re-estimates the adjusted distance and time differences by group. Conditional on public-transport use, the gaps in distance and time across gender–marital status groups essentially disappear: married women, married men, single women, and single men commute very similar distances and spend very similar amounts of time traveling. This indicates that the headline distance and time gaps are not driven by differences in commuting behavior conditional on mode but rather by differences in who has access to private modes of transport.

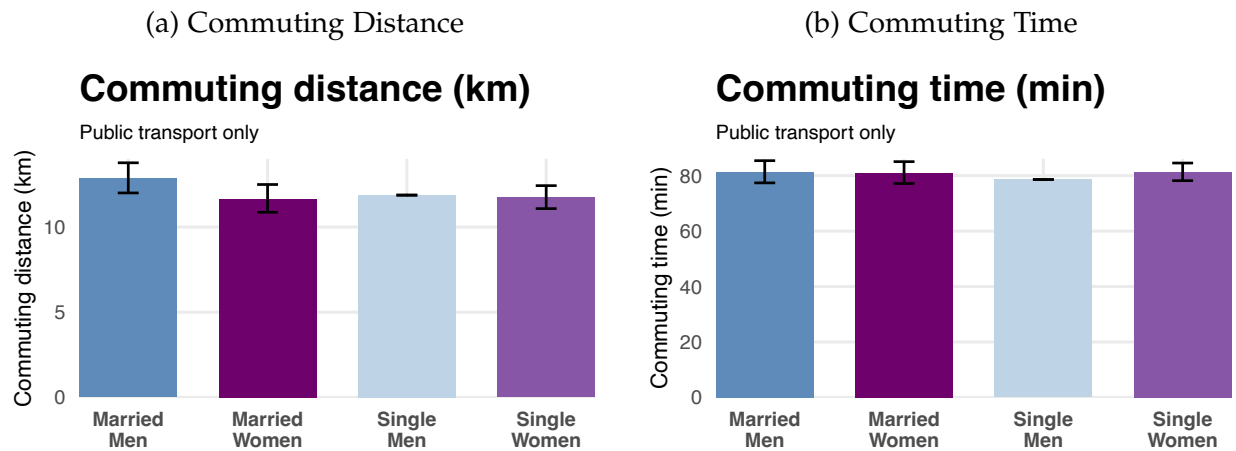
Living near transport infrastructure links. A third cut looks at workers who live close to major transport links (rail stations and BRT corridors). Figure A12 reports the adjusted commuting outcomes for this subsample. For individuals living within 750m from high-capacity public transport, such as bus corridors and subways, the gender–marital gaps in distance and time disappear, despite there still being a substantial difference in the

Figure A10: Commuting Outcomes Conditional on Residence and Workplace Fixed Effects



Notes: The figure replicates Figure 3, additionally absorbing residence and workplace zone fixed effects. Predicted commuting outcomes are calculated as the baseline mean for single men plus the estimated regression effects for each group, with controls for age, education, wealth, and household composition. Confidence intervals are derived from the regression estimates. Identification therefore comes from comparing individuals who live in the same residence zone and work in the same workplace zone.

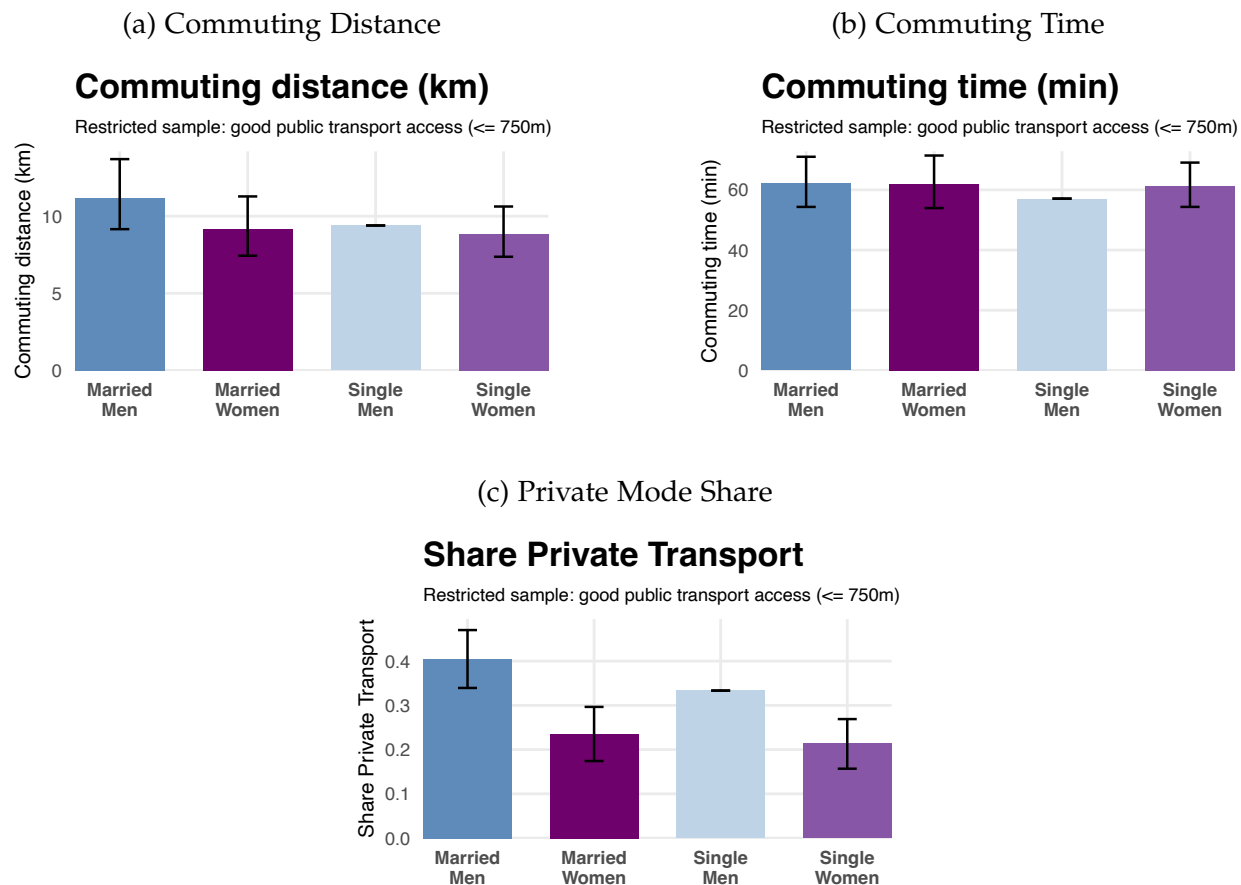
Figure A11: Commuting Outcomes Conditional on Public Transport Use



Notes: The figure replicates the distance and time panels of Figure 3 restricting the sample to workers who commute by public transport. Predicted commuting outcomes are calculated as the baseline mean for single men plus the estimated regression effects for each group, with controls for age, education, wealth, household composition, and residence location bin fixed effects. Confidence intervals are derived from the regression estimates. Conditional on public-transport use, the gender-marital gaps in commuting distance and time are essentially zero.

private-mode share. When fast public transport is available close to home, not having access to vehicles has a much smaller impact in women's capacity to cover the same distance as men in the same time frame. Together with the previous two exercises, this points to a consistent reading of the data: the gender-marital differences in commuting outcomes are not driven by residential or workplace sorting, but rather operate through mode choice and through unequal access to fast public transport.

Figure A12: Commuting Outcomes for Workers Living Near Major Transport Links



Notes: The figure replicates Figure 3 restricting the sample to workers who live within 750m to major public transport infrastructure (rail stations and BRT corridors). Predicted commuting outcomes are calculated as the baseline mean for single men plus the estimated regression effects for each group, with controls for age, education, wealth, household composition, and residence location bin fixed effects. Confidence intervals are derived from the regression estimates.

B Model Appendix

This Appendix Section summarizes how the model in Section 3 extends to include singles and how their entry modifies general equilibrium. The main changes operate through residential floor space demand, which now accounts for the demand of single people as well.

B.1 Preferences of Singles

A single household consists of one individual of gender $g \in \{m, f\}$ who chooses residence n and workplace i , where $i = 0$ denotes home production. Let $\phi_i \in \{0, 1\}$ indicate participation in home production. Utility is Cobb–Douglas in consumption c_{ni} and residential floor space h_{ni} , scaled by location amenities B_{ni}^{sg} and discounted by iceberg commuting costs d_{ni}^{sg} :

$$U_{ni}^{sg}(\omega) = \frac{B_{ni}^{sg} z_{ni}^{sg}(\omega)}{d_{ni}^{sg}} \left(\frac{c_{ni}}{\beta^s} \right)^{\beta^s} \left(\frac{h_{ni}}{1 - \beta^s} \right)^{1 - \beta^s}, \quad \beta^s < 1. \quad (\text{B.1})$$

Iceberg commuting costs rise exponentially with travel time t_{ni}^{sg} : $d_{ni}^{sg} = e^{\kappa^{sg} t_{ni}^{sg}} \geq 1$, where κ^{sg} captures individual-specific commuting decay. Singles maximize utility subject to

$$c_{ni} + q_{Rn} h_{ni} \leq (1 - \tau) w_i^{sg} + \phi_i T, \quad (\text{B.2})$$

with residential rent q_{Rn} and transfer T received when not earning firm wages. Conditional demands take the standard Cobb–Douglas form:

$$c_{ni} = \beta^s HI_i^s, \quad h_{ni} = \frac{1 - \beta^s}{q_{Rn}} HI_i^s, \quad (\text{B.3})$$

where single-household income is $HI_i^s = (1 - \tau) w_i^{sg} + \phi_i T$.

B.2 Residence and Workplaces Decision

Singles choose residence n and workplace i to maximize utility. With Fréchet-distributed shocks z_{ni}^{sg} , the closed-form location share for singles is:

$$\pi_{ni}^{sg} = \frac{(B_{ni}^{sg} HI_i^s)^{\epsilon^s} (d_{ni}^{sg} q_{Rn}^{1 - \beta^s})^{-\epsilon^s}}{\sum_{r=1}^N \sum_{p=0}^N (B_{rp}^{sg} HI_p^s)^{\epsilon^s} (d_{rp}^{sg} q_{Rr}^{1 - \beta^s})^{-\epsilon^s}} \equiv \frac{L_{ni}^{sg}}{L^{sg}}, \quad (\text{B.4})$$

where L_{ni}^{sg} is the number of singles of gender g living in n and commuting to i , and L^{sg} is the total number of singles of gender g . Summing over all workplace locations provides the residence share:

$$\pi_{Rn}^{sg} = \frac{\sum_{p=0}^N \left(B_{np}^{sg} H I_p^s \right)^{\epsilon^s} \left(d_{np}^{sg} q_{Rn}^{1-\beta^s} \right)^{-\epsilon^s}}{\sum_{r=1}^N \sum_{p=0}^N \left(B_{rp}^{sg} H I_p^s \right)^{\epsilon^s} \left(d_{rp}^{sg} q_{Rr}^{1-\beta^s} \right)^{-\epsilon^s}} \equiv \frac{L_{Rn}^{sg}}{L^{sg}}, \quad (\text{B.5})$$

and singles sort across the city based on location amenities, commuting costs, income and housing prices. The labor share of singles of gender g at each location is obtained summing over all the residence locations:

$$\pi_{Fi}^{sg} = \frac{\sum_{r=1}^N \left(B_{ri}^{sg} H I_i^s \right)^{\epsilon^s} \left(d_{ri}^{sg} q_{Rr}^{1-\beta^s} \right)^{-\epsilon^s}}{\sum_{r=1}^N \sum_{p=0}^N \left(B_{rp}^{sg} H I_p^s \right)^{\epsilon^s} \left(d_{rp}^{sg} q_{Rr}^{1-\beta^s} \right)^{-\epsilon^s}} \equiv \frac{L_{Rn}^{sg}}{L^{sg}}, \quad (\text{B.6})$$

B.3 Commuter Market Clearing

Labor supply from singles must be consistent with commuting allocations:

$$L_{Fi}^{sg} = \sum_{n=1}^N \pi_{ni|n}^{sg} L_{Rn}^{sg}, \quad (\text{B.7})$$

ensuring that the mass of workers sg at workplace i equals the residential mass willing to commute there. Expected single-household income at residence n is:

$$v_n^{sg} = \sum_{i=0}^N \pi_{ni|n}^{sg} H I_i^s, \quad (\text{B.8})$$

which enters equilibrium floor space prices and firm labor demand through its effect on aggregate real income.

B.4 Floor Space Market Clearing

Including singles augments both sides of the floor space market clearing. Firm floor space prices now depend on total labor supplied, including singles:

$$q_{Fi} = \frac{1 - \alpha}{\alpha} \frac{\tilde{w}_i \left(L_{Fi}^c + \sum_g L_{Fi}^{sg} \right)}{H_{Fi}}, \quad (\text{B.9})$$

while residential housing prices reflect combined housing demand from couples and singles:

$$q_{Rn} = \frac{1 - \beta^s}{H_{Rn}} \left(v_n^c L_{Rn}^c + \sum_g v_n^{sg} L_{Rn}^{sg} \right). \quad (\text{B.10})$$

Thus, adding singles increases competition for housing, and shifts workplace allocations and real income expectations.

B.5 General Equilibrium

Given the model parameters $\{\alpha, \beta^s, \epsilon^c, \epsilon^{sg}, \kappa^{og}, \gamma^{og}\}$, city population L^{og} , government cash transfers T , and location characteristics $\{A_i, B_{Ri}^c, B_{Ri}^{sg}, B_{Fi}^{og}, B_{n0}^{og}, t_{ni}^{og}\}$, the general equilibrium is represented by the vector of endogenous objects $\{L_{Ri}^c, L_{Ri}^{sg}, L_{Fi}^c, L_{Fi}^{sg}, \tilde{w}_i, q_{Fi}, q_{Ri}, \tau\}$ determined by the following equations:

1. Residential population equation for couples (10) and singles (B.5).
2. Labor supply equation of couples (11) and singles (B.6).
3. Unit cost condition and utility maximization from firms (18).
4. Floor space demand from firms (B.9).
5. Floor space demand from residences (B.10).
6. Government budget constraint (25).

C Estimation Details

This appendix details the estimation and inversion procedures used to quantify the model in Section 4. The objective is to make explicit (i) the gravity equations used to identify commuting frictions and participation amenities, (ii) the inversion steps used to recover structural fundamentals, and (iii) the numerical algorithms used to solve the model.

I partition model objects into four groups. First, externally calibrated objects are $(\alpha, \beta^c, \beta^s, s^{og}, \varphi, T)$, using expenditure shares, wage regressions, and institutional assumptions as described in the main text. Second, estimated reduced-form objects are the commuting semi-elasticities ν^{og} and participation fixed effects from gravity equations. Third, estimated structural elasticities are (ϵ^c, ϵ^s) , recovered within model inversion from wage-dispersion moments. Fourth, inverted fundamentals are workplace amenities $\{B_{Fi}^{og}\}$, residential amenities $\{B_{Rn}^o\}$, productivities $\{A_i\}$, and floor-space objects.

The inversion uses observed residence masses, commuting flows, wages, and floor-space prices, together with the model restrictions in Section B. Identification is up to scale normalizations, implemented as geometric-mean normalizations on inverted amenity and productivity terms.

C.1 Externally calibrated parameters

Housing expenditure share I calibrate housing expenditure shares using the Brazilian Household Budget Survey (POF 2017–2018), with expenditure and income defined at the level of the consumption unit.²⁹ Housing expenditure is annualized rent (observed and imputed), divided by annualized consumption-unit income. I estimate weighted regressions of that share on indicators for couple and single, controlling for unit size and state fixed effects, in the urban sample. The model targets the average predicted shares for couple and single types, which implies $\beta_h^c = 0.20$ and $\beta_h^s = 0.25$ in the baseline quantification.

²⁹In IBGE terminology, a consumption unit (unidade de consumo) comprises people who live together and pool resources for food and other common expenses. It need not coincide with the dwelling (domicílio): one address can contain several consumption units when budgets are separate. For the couple and single types used in the model, the consumption unit typically matches the model type. Shares use consumption-unit total income in the denominator; individual-level fields enter only where needed for type composition.

Floor space share in production The production floor-space share is set to $1 - \alpha = 0.2$ (Tsivanidis, 2022).

Government transfer and home production productivity Two objects govern the outside option of home production: the transfer T and home productivity φ . In the quantitative implementation, these terms enter additively in household income when an individual is out of paid work, so I calibrate their sum ($T + \varphi$) to a fixed fraction of the 2007 Brazilian minimum wage. Within this structure, the exact decomposition between transfer income and home-production return is not separately identified for baseline inversion; what matters for labor-supply choices is their combined level. A higher calibrated value of $(T + \varphi)$ strengthens the outside option, shifts the extensive margin toward home production, and, through the government budget constraint, requires a higher flat labor-income tax to finance transfers.

Differences across types are then captured by type-specific participation amenities, which can be interpreted as variation in social norms and other non-pecuniary determinants of home production.

C.2 Gravity equation details

From choice probabilities to estimating equation Starting from the couples' conditional choice probabilities, the implied flow equation for residence-workplace-workplace cells (n, i, j) can be written as:

$$L_{nij}^c \propto \left(B_{n0}^{cm} B_{n0}^{cf} \right) \left(B_{Rn}^c q_n^{\beta^c - 1} \right) \left(B_{Fi}^{cm} B_{Fj}^{cf} H I_{ij}^c \right) \exp \left(v^{cm} t_{ni}^{cm} + v^{cf} t_{nj}^{cf} \right), \quad (\text{C.1})$$

where $v^{og} = \epsilon^o \kappa^{og}$ for each group. Taking logs and collecting terms yields:

$$\log L_{nij}^c = \eta_{n0}^{cm} + \eta_{n0}^{cf} + \delta_n + \zeta_{ij} + v^{cm} t_{ni}^{cm} + v^{cf} t_{nj}^{cf} + u_{nij}, \quad (\text{C.2})$$

with η_{n0}^{cg} origin-home-production fixed effects, δ_n residence fixed effects, and ζ_{ij} workplace-pair fixed effects.

Estimation and endogeneity correction Equation (C.2) is estimated by PPML to accommodate sparse bilateral flow matrices and zero-flow cells. Travel times may be endogenous to commuting flows due to equilibrium transport conditions. I therefore use a control-function strategy: first-stage residualized travel times are obtained using distance instruments and high-dimensional fixed effects, and second-stage PPML includes

the first-stage residuals as control functions. In practice, the IV-control-function specification yields commuting coefficients with larger absolute magnitude than non-IV specifications, suggesting that attenuation from endogenous travel times downward biases the PPML estimates. Once this source of endogeneity is addressed, commuting disutility is stronger, implying that the governments disproportionately invest in high-demand bilateral flows, compressing travel times precisely where flows are large anyway.

Mapping to structural objects Estimated commuting coefficients (ν^{cm}, ν^{cf}) feed directly into the inversion. Estimated origin-home-production fixed effects are exponentiated to construct participation amenities by residence and worker type. The same logic applies to singles with a standard origin-destination gravity equation and type-specific commuting semi-elasticities.

C.3 Model inversion of fundamentals

The inversion proceeds sequentially. Let observed residence masses be $\{L_n^o\}$, observed commuter masses be $\{L_{ni}^{og}\}$ or $\{L_{nij}^c\}$, and observed wages and prices be $\{w_i^{og}, q_n, Q_i\}$.

Observed wages enter in two layers. First, for each location, I compute the average wages from married men and normalize it. Second, group-specific wages are constructed by multiplying this base wage by estimated discrimination wedges $(1 - \gamma^{og})$ (from equation (17) in the main text). This “wage wedge” captures persistent differences across gender-marital-status groups and ensures that inversion separates type-specific wage components from location fundamentals.

Step 1: transformed wage objects from commuter clearing For each group, I solve commuter market-clearing equations for transformed wage terms ω such that model-implied workplace masses match observed workplace masses, conditional on observed residence masses and commuting frictions.

Step 2: estimation of the Fréchet dispersion parameter ϵ For each group, I estimate ϵ from the mapping between transformed wages ω and implied wages $\tilde{w} = \omega^{1/\epsilon}$. Let w^{obs} be observed wages (normalized to the same support as transformed wages). The estimator chooses ϵ to match the dispersion of log wages:

$$\hat{\epsilon} = \arg \min_{\epsilon \in [\underline{\epsilon}, \bar{\epsilon}]} \left[\text{Var}(\log w^{obs}) - \text{Var}(\log (\omega^{1/\epsilon})) \right]^2. \quad (\text{C.3})$$

This is implemented separately for singles and couples, with group-specific transformed-wage objects and common constrained optimization settings. The objective in (C.3) is solved by constrained nonlinear optimization with box bounds. Missing cells are aligned between observed and transformed wage vectors before moment computation. Higher ϵ implies lower idiosyncratic dispersion and stronger sensitivity of choice probabilities to systematic utility differences.

Step 3: adjusted wages using estimated ϵ Given estimated ϵ^o , adjusted wages are:

$$\tilde{w}_i^{og} = (\omega_i^{og})^{1/\epsilon^o}. \quad (C.4)$$

For couples, the object is recovered at the workplace-pair level and mapped back to individual workplace components using model structure.

Step 4: workplace amenities Holding adjusted wages and commuting costs fixed, workplace amenities $\{B_{Fi}^{og}\}$ are recovered from conditional workplace-choice equations so that predicted conditional commuting shares match observed commuting shares by residence. Intuitively, workplace amenities absorb systematic differences in workplace attractiveness that are not explained by wages or commuting costs. They are estimated separately by worker type because the same location can offer different effective non-pecuniary value across groups (for example, due to sectoral composition, schedule compatibility, or workplace-specific constraints that differ by gender and household structure). A higher B_{Fi}^{og} means that, conditional on wages and travel times, type *og* workers are disproportionately likely to commute to location *i*.

Step 5: residential amenities Residential amenities $\{B_{Rn}^o\}$ are recovered from residence-choice equations:

$$L_n^o \propto \left(B_{Rn}^o v_n^o q_n^{\beta^o - 1} \right)^{\epsilon^o}, \quad (C.5)$$

where v_n^o is expected income at residence *n*. Rearranging this expression gives:

$$B_{Rn}^o \propto (L_n^o)^{1/\epsilon^o} (v_n^o)^{-1} (q_n)^{1-\beta^o}, \quad (C.6)$$

up to a multiplicative constant. In implementation, I apply a sequence of normalizations to isolate relative residential attractiveness: (i) residence shares are normalized by their

geometric mean, (ii) price and commuting-access terms are normalized analogously, and (iii) the recovered amenity vector is re-scaled so that $\text{geomean}(B_R^o) = 1$. This procedure preserves all cross-location variation and only removes the non-identified level.

Step 6: productivity Using the firm block and observed commercial prices and wages, productivity is inverted location by location:

$$A_i = \left(\frac{Q_i}{1 - \alpha} \right)^{1-\alpha} \left(\frac{w_i}{\alpha} \right)^\alpha. \quad (\text{C.7})$$

I then apply a geometric-mean normalization to $\{A_i\}$.

Step 7: floor-space objects and implied quantities Given (A_i, w_i, Q_i) and residence-side objects, I recover commercial and residential floor-space demands and corresponding implied supply shifters used in the equilibrium stage. This completes the baseline structural decomposition.

C.4 Numerical algorithms

The routines below implement the objects described in Section C.3. The first two are *fixed-point inversions*: each searches for wage indices that align the model’s predicted workplace employment with the data, *conditional on* the residence distribution, travel costs, participation amenities, and the other inputs assigned to that stage, including prices and fundamentals, which are held fixed for the subroutine. They are best thought of as commuting market clearing–accounting identities that link residence choices to workplace totals at a steady state implied by those inputs. The third routine is an outer fixed point that solves for general equilibrium and finds the steady state of wages, residential and commercial floor-space prices, and populations, conditional on fundamentals. In the equilibrium solver loop, level shifters provide a way to stabilize and target means—matching average wages, residential and commercial prices, and home-production shares—while the high-dimensional fundamentals govern spatial heterogeneity; freezing them in counterfactuals ensures that the exercises do not implicitly re-fit a baseline scale normalization.

Transformed-wage fixed point (inversion Step 1) The unknowns are the transformed wage indices ω that enter workplace choice probabilities before ϵ is used to rescale them into money wages. Conditional on observed residence populations, commuting costs $\exp(\nu^{og} t_{ni}^{og})$, and participation amenities, each iteration does the following. First,

for each worker type, I build residence-to-workplace conditional probabilities λ_{ni}^{og} (row-normalized kernels). Second, I aggregate predicted workplace employment by summing those probabilities against residence masses. Third, I rescale each workplace component of ω by the ratio of observed to predicted employment, renormalize to geometric mean one, and apply a damped update so the iteration is stable in sparse cells. The loop stops when a rounded employment-fit criterion falls below a small fraction of total predicted employment. This block corresponds exactly to inversion Step 1; it delivers the ω that feeds the ϵ objective in Step 2.

Fréchet ϵ (inversion Step 2) In Step 2, I minimize the squared difference between the variance of log observed wages and the variance of log implied wages $\omega^{1/\epsilon}$ over a bounded interval, using a constrained nonlinear solver. The transformed wages ω are held fixed at the Step 1 solution while ϵ moves.

Adjusted-wage fixed point (inversion Step 3) Once $\hat{\epsilon}$ is fixed from Step 2, I solve again for commuting consistency, but now on the objects that enter utility as money wages: adjusted wages satisfy $\omega = \tilde{w}^\epsilon$ (implemented by iterating on ω with the same employment-updating map as above, started from $\tilde{w}^\epsilon$). The economic content is unchanged—still matching observed workplace masses—but the rescaling links the fixed point to Step 3 in Section C.3. Singles and couples are handled separately because their choice sets and tensors of workplace pairs differ.

Outer equilibrium loop (post-inversion) After amenities, productivity, and floor-space objects are recovered, the equilibrium solver searches for a vector of wages, residential floor-space prices, commercial floor-space prices, and implied residence and workplace populations that are mutually consistent with the model’s choice probabilities, production side, and housing demand. One outer iteration proceeds as follows. First, given current guesses (with optional log level shifters on wages and prices), I evaluate the same Fréchet kernels as in inversion—now using adjusted wages, full amenities, and housing terms—and obtain predicted residence masses, workplace masses, and home-production shares. Second, I update workplace output and wages from the production function and efficiency units. Third, I update residential and commercial floor-space prices from residential spending needs and firm revenue, again respecting floor-space supply splits. Fourth, in calibration mode only, I nudge the level shifters so that mean wages and prices (and optionally home-production shares) track their data counterparts;

in counterfactual mode those shifters are frozen. Finally, I damp the update to the endogenous wage and price vectors and test convergence on the maximum absolute change across wages and both price types. The loop terminates when that sup-norm falls below a tight tolerance or a maximum iteration cap binds.

Numerical stability Throughout, I treat missing or zero cells in employment matrices as inactive in the updating step, keep small floors in ratios to avoid division by zero, and use geometric-mean normalizations to fix scales that the theory leaves indeterminate.